

AI Clones and Creations

Annie Liang
(Northwestern)

IDSS

Introduction

Generative AI systems are increasingly used to produce new creative content.

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Why this is qualitatively new:

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- Outputs can be rich and complex—**cartoons**

Generating New Yorker-Style Cartoons with AI

Harnessing the power of OpenAI's ChatGPT to Transform Cartoon ideas into Reality



Georgiy Ermakov

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5 min read · Oct 25, 2023

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- Outputs can be rich and complex—cartoons, **novels**

The AI-generated Books Trend is Getting Worse

It has gotten so bad that Amazon's new policy limits self-published Kindle ebooks to three a day.



Sascha Brodsky, Contributor
January 24, 2024

🕒 4 Min Read

AI BUSINESS

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Tilly Norwood: how scared should we be of the viral AI ‘actor’?

A bunch of code is being pushed as the next Scarlett Johansson, a creation that is already causing pushback from real human actors

Tilly Norwood



A 2025 AI-generated image of Tilly Norwood, taken from the character's [Instagram](#) account

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AI Agents Simulate 1,052 Individuals' Personalities with Impressive Accuracy

Stanford HCAI, 2025

It's no longer science fiction: Your personality – your beliefs, quirks, and decision-making patterns – can be captured and brought to life inside an artificial mind.

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AI Training

Upload your resume and provide context about your experience, goals, and preferences



Clone Creation

Our AI creates an intelligent representation that understands your professional profile



Smart Matching

Your AI clone engages with recruiters, answering questions and showcasing your skills

Introduction

AI clones and creations will reshape:

- How people interact with one another
- What and how humans create

They will transform institutions and markets, such as

- Matching markets
- Markets for creative work

We need to think through their welfare consequences and how to regulate them, e.g., how to allocate creative ownership and rights.

These issues are **already** present given existing AI technology, and will grow even more so as this technology improves.

This Talk

Will discuss two aspects of this through two papers:

- Artificial Intelligence Clones, 2026

What are the welfare implications of the use of AI clones for searching/matching over human candidates?

- Creative Ownership in the Age of AI (joint with Jay Lu), 2026

How should we define and protect intellectual property in a post-generative AI world?

Artificial Intelligence Clones

Annie Liang
(Northwestern)

Introduction

LLMs, fine-tuned on our personal data, may soon be able to mimic individual personalities (“AI clones,” “AI agents,” “AI delegates”)

This technology has the potential to transform how we search over human candidates, in particular in settings where direct evaluation is costly.

Labor and dating markets are already moving in this direction.

REPLICANT

AI-powered candidate replication platform helping job seekers create intelligent digital representatives for recruitment conversations.



AI Training

Upload your resume and provide context about your experience, goals, and preferences



Clone Creation

Our AI creates an intelligent representation that understands your professional profile



Smart Matching

Your AI clone engages with recruiters, answering questions and showcasing your skills

Ready to Get Started?

Join the future of recruitment and let your AI clone work for you.

Create Your AI Clone

Ribbon raised \$8.2M to fix hiring globally >

Meet your AI-powered recruiter

Ribbon talks with every candidate, scores their answers, and shares insights that help you hire faster and smarter.

Book a demo

Start for free



Daphne D.
Operations Manager

This was my first time doing an interview like this and it was so smooth!



Hassan J.
DevOps Engineer

I like the new AI approach, and I thought the agent was very interactive and smart and responded well to changing dynamics of the conversation!



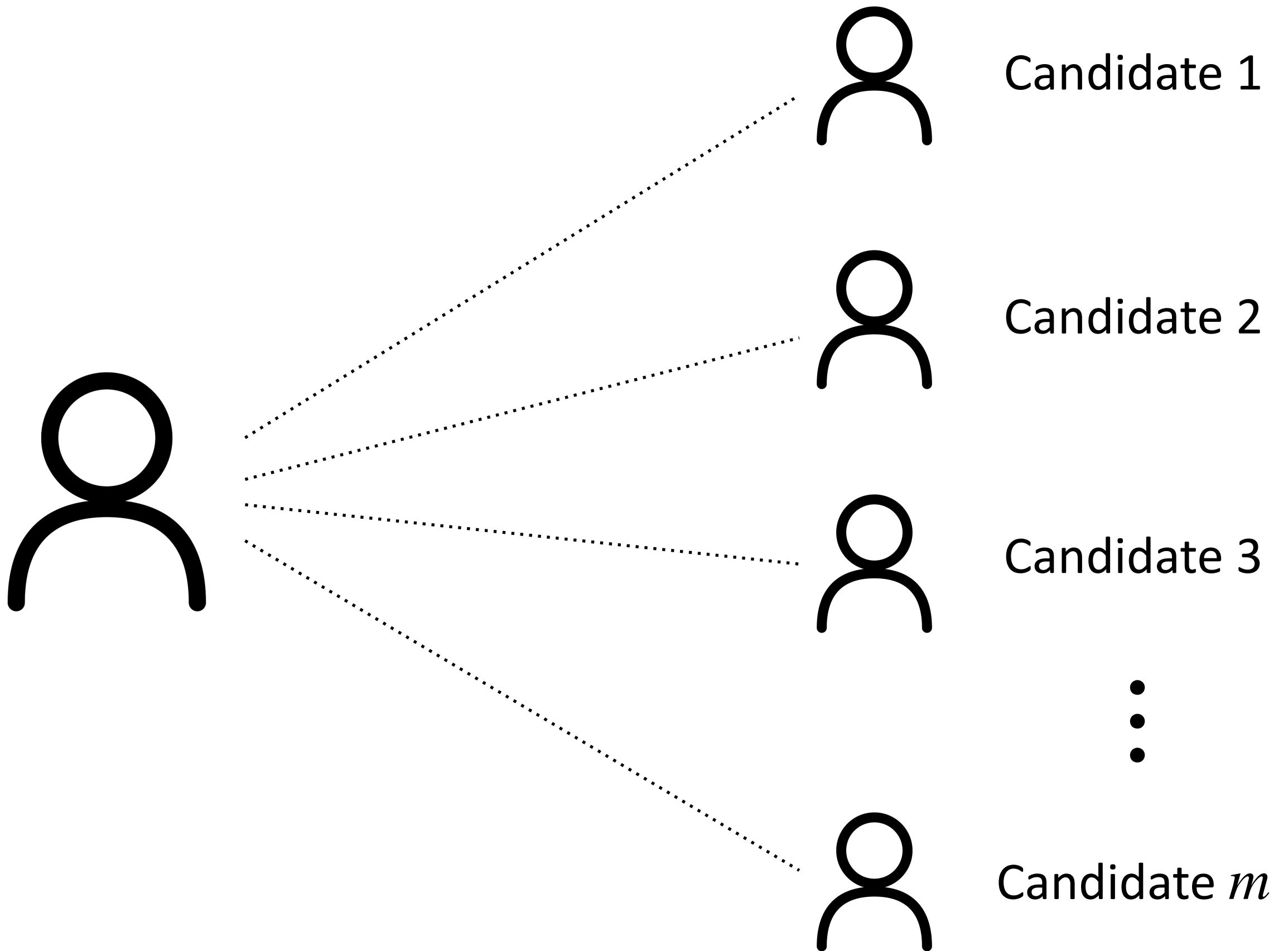
Jamie K.

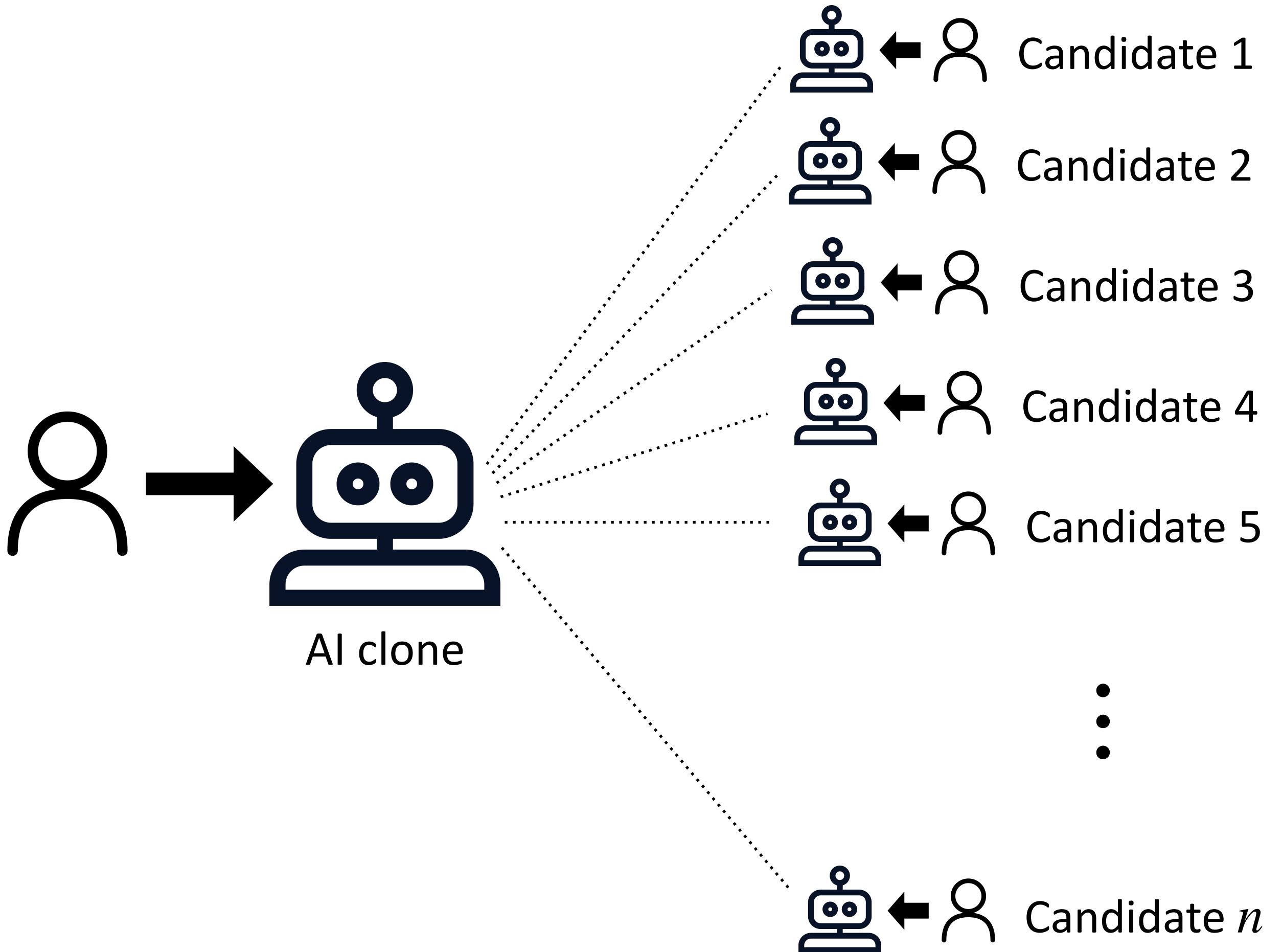
Are A.I. Clones the Future of Dating? I Tried Them for Myself.

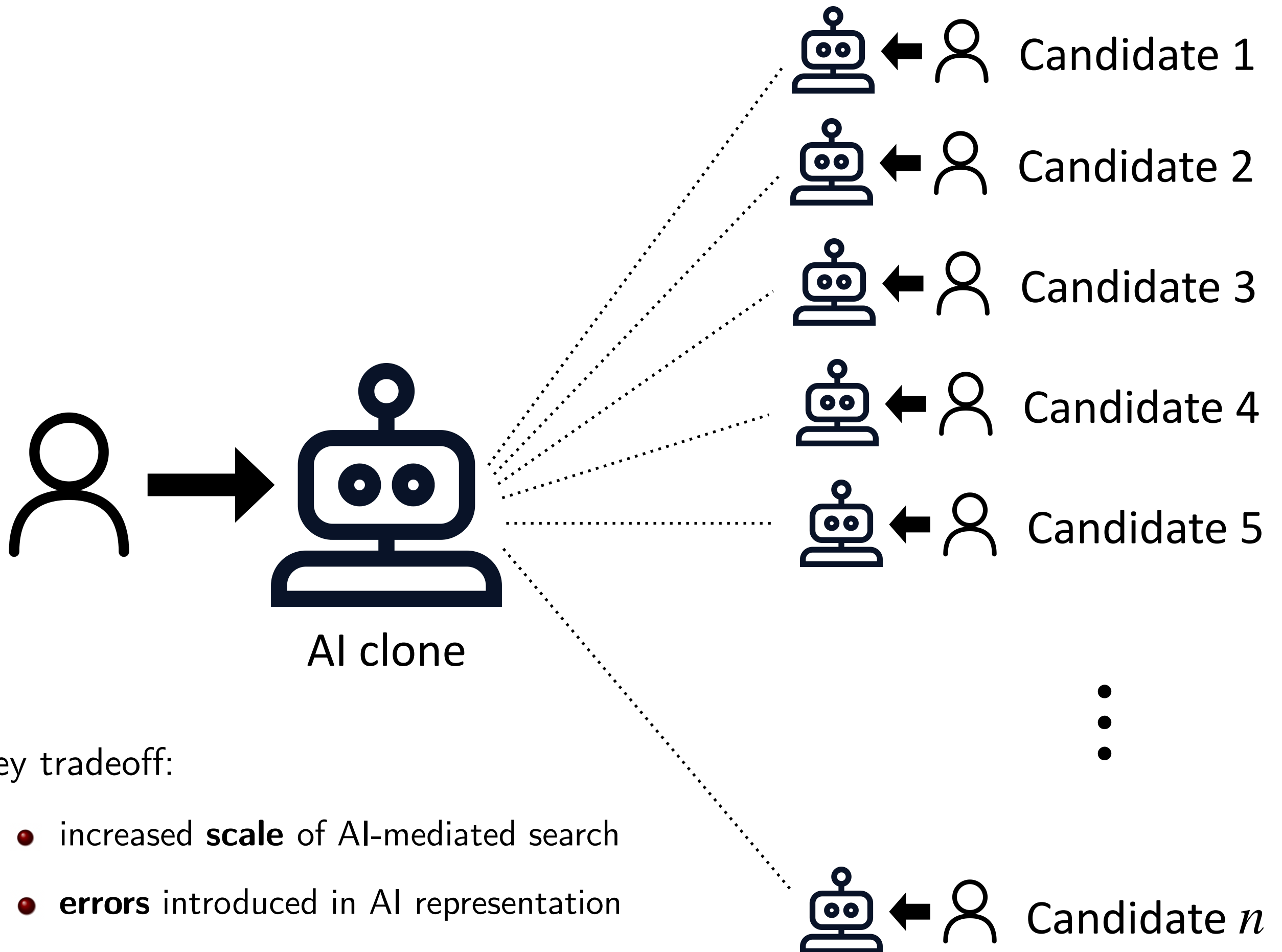
executives in the dating app industry have begun pitching a future in which people can create A.I. clones of themselves that date other clones and relay the results back to their human counterparts.

Dating Apps Suck. A.I. Clones Are Making Them Even Weirder.

‘My clone is on its way to date all of San Francisco in one month.’







This Paper

- Individual personalities are modeled as points in k -dimensional Euclidean space, and their AI clones are modeled as noisy approximations of their personalities
- Compare two search regimes:
 - **In-person search**—randomly meet some number of individuals and match to the most compatible among them
 - **AI-mediated search**—match to the person with the most compatible AI clone

This Paper

- Individual personalities are modeled as points in k -dimensional Euclidean space, and their AI clones are modeled as noisy approximations of their personalities
- Compare two search regimes:
 - **In-person search**—randomly meet some number of individuals and match to the most compatible among them
 - **AI-mediated search**—match to the person with the most compatible AI clone
- **AI-equivalent sample size**: smallest m such that search over m people in person is better than participating on an AI platform with infinite candidates
- Main results
 - The AI-equivalent sample size is finite (i.e., value of AI platform is **capped**)
 - When people are sufficiently high dimensional (k is large), the AI-equivalent sample size is 2
- Takeaway: **the complexity of match compatibility matters** for how we should think about human vs. AI search

Setup

Individuals are represented by vectors in $\mathbb{R}^k$

- each entry is an attribute of the person

possible attributes:

extraversion

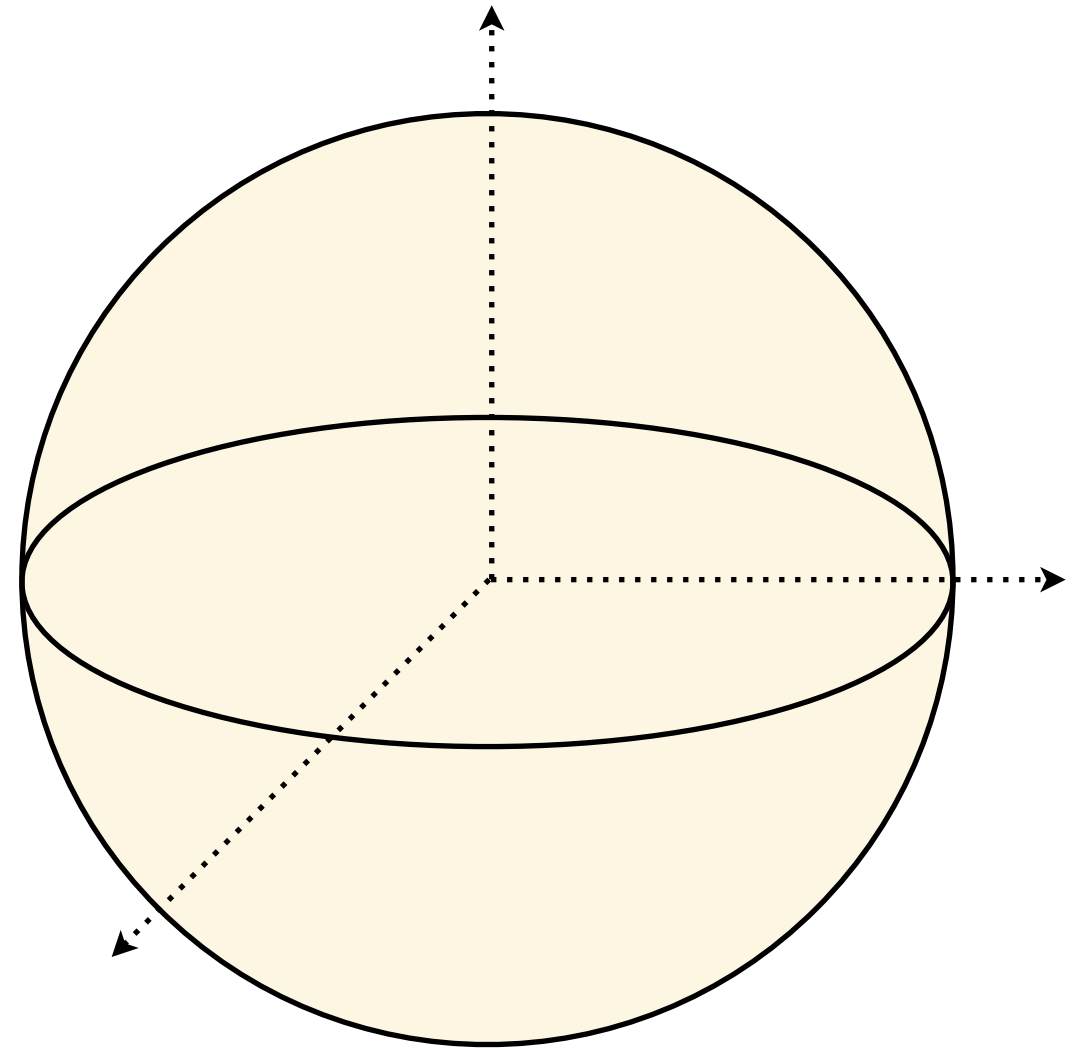
intellectual curiosity

relationship to time

style of humor

relationship to authority

speed of decision-making

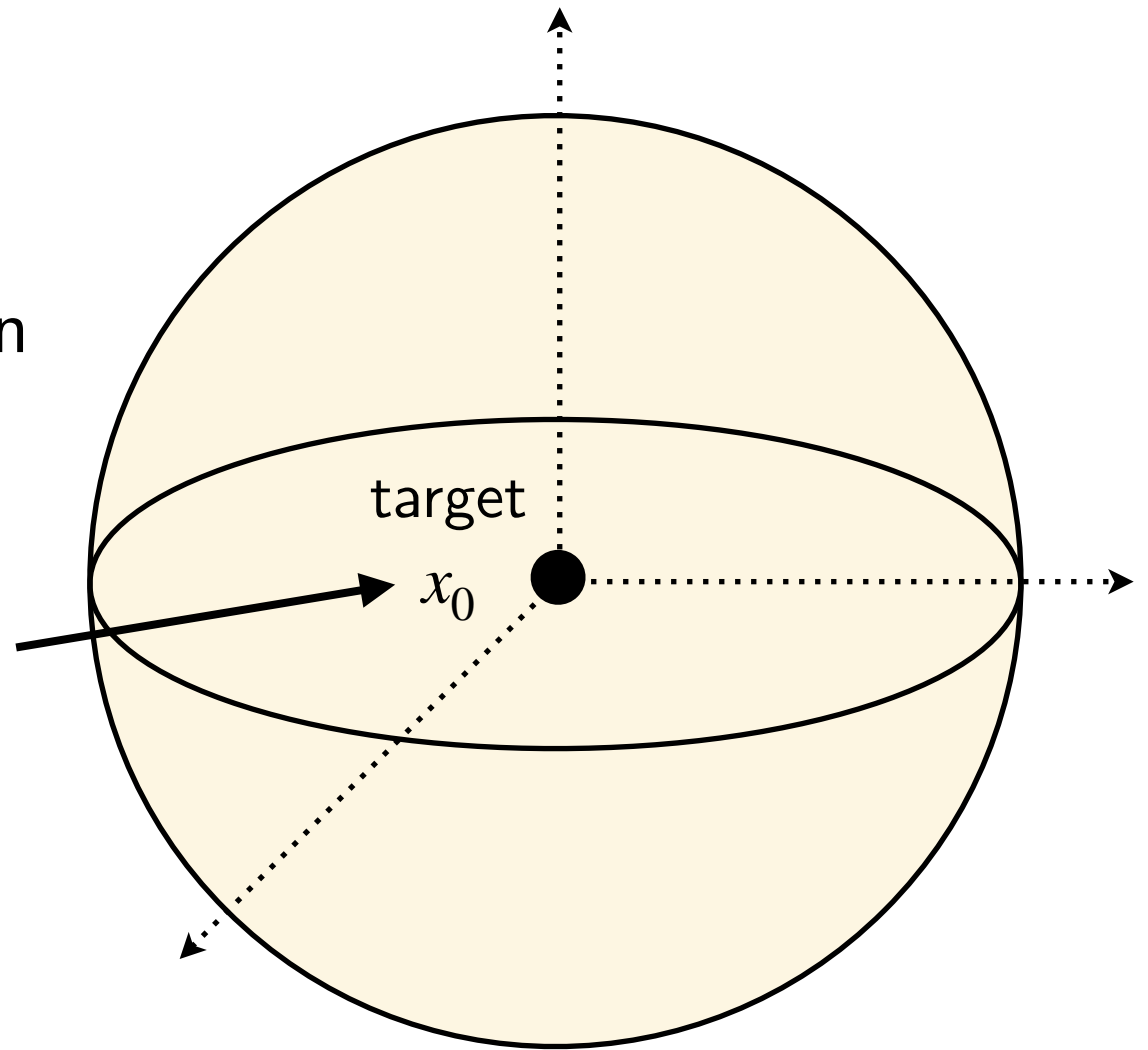


Setup

Individuals are represented by vectors in $\mathbb{R}^k$

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The searcher's ideal match is normalized to the origin and denoted x_0 (the **target**)



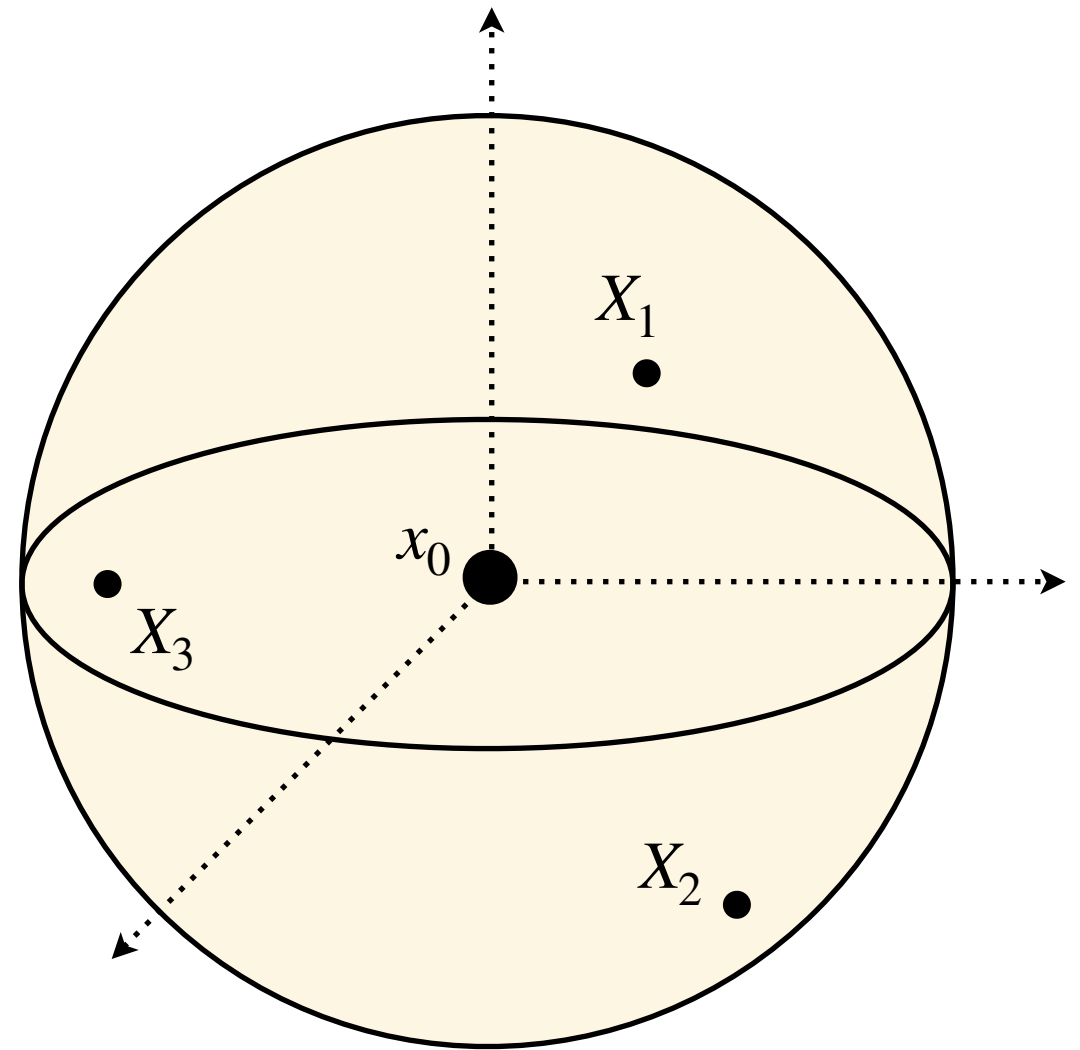
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People in the population $X_i \in \mathbb{R}^k$ are drawn uniformly at random from the unit ball B_k



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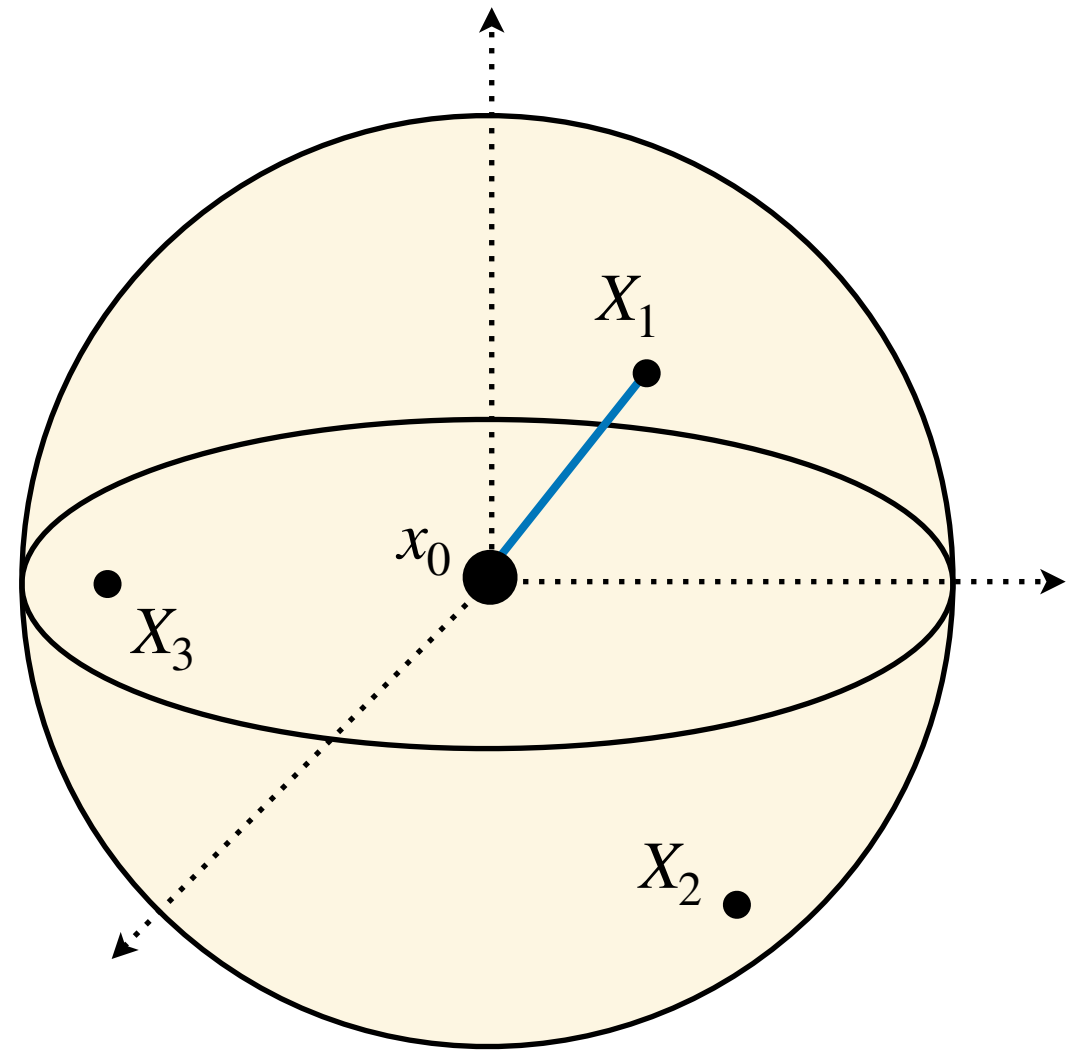
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The searcher receives utility $-\|X_i\|$ from being matched to individual i

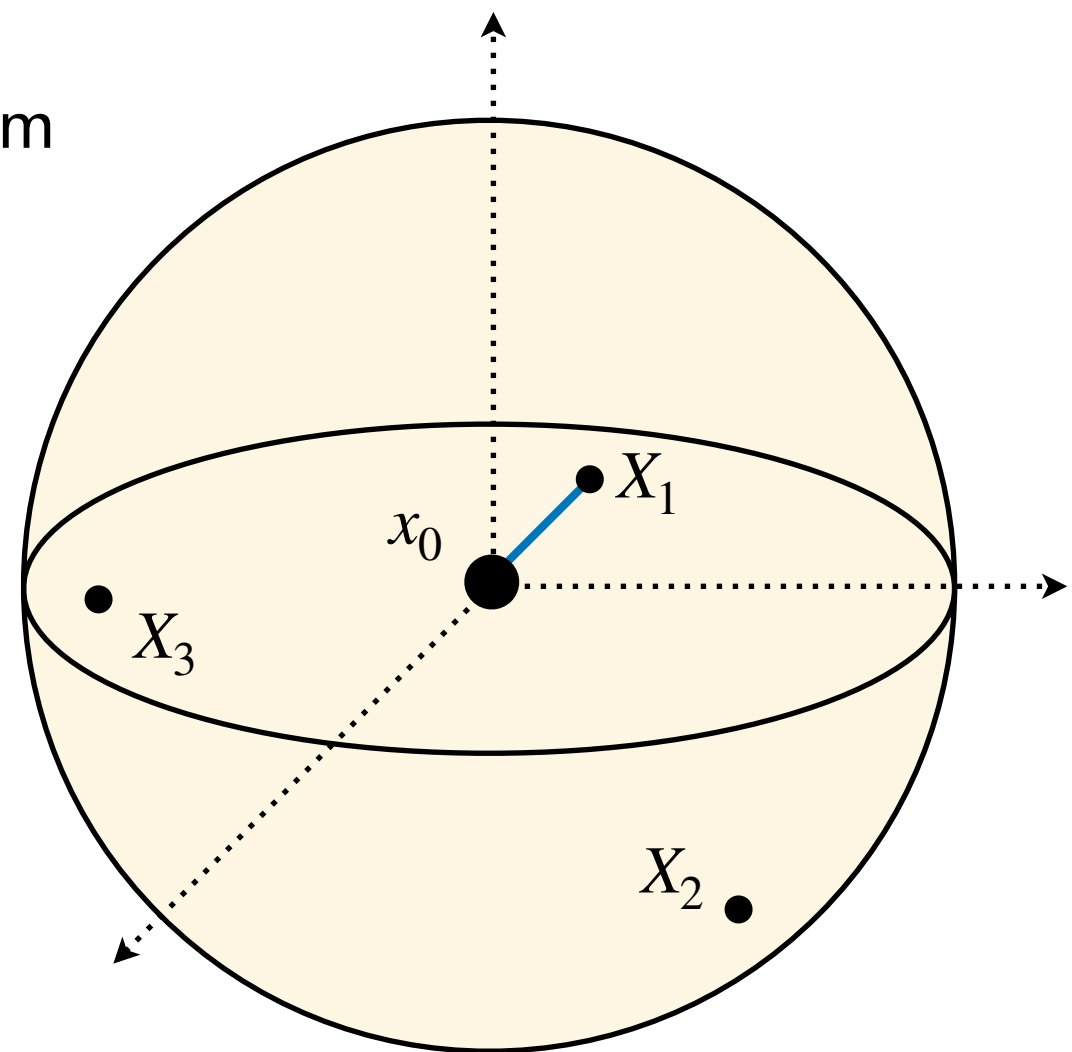


In-Person Search

Under **in-person** search,

- The searcher meets m individuals $X_1, \dots, X_m$ draw i.i.d. from a uniform distribution on the unit ball
- Matches to the most compatible among them
- The searcher's expected utility is

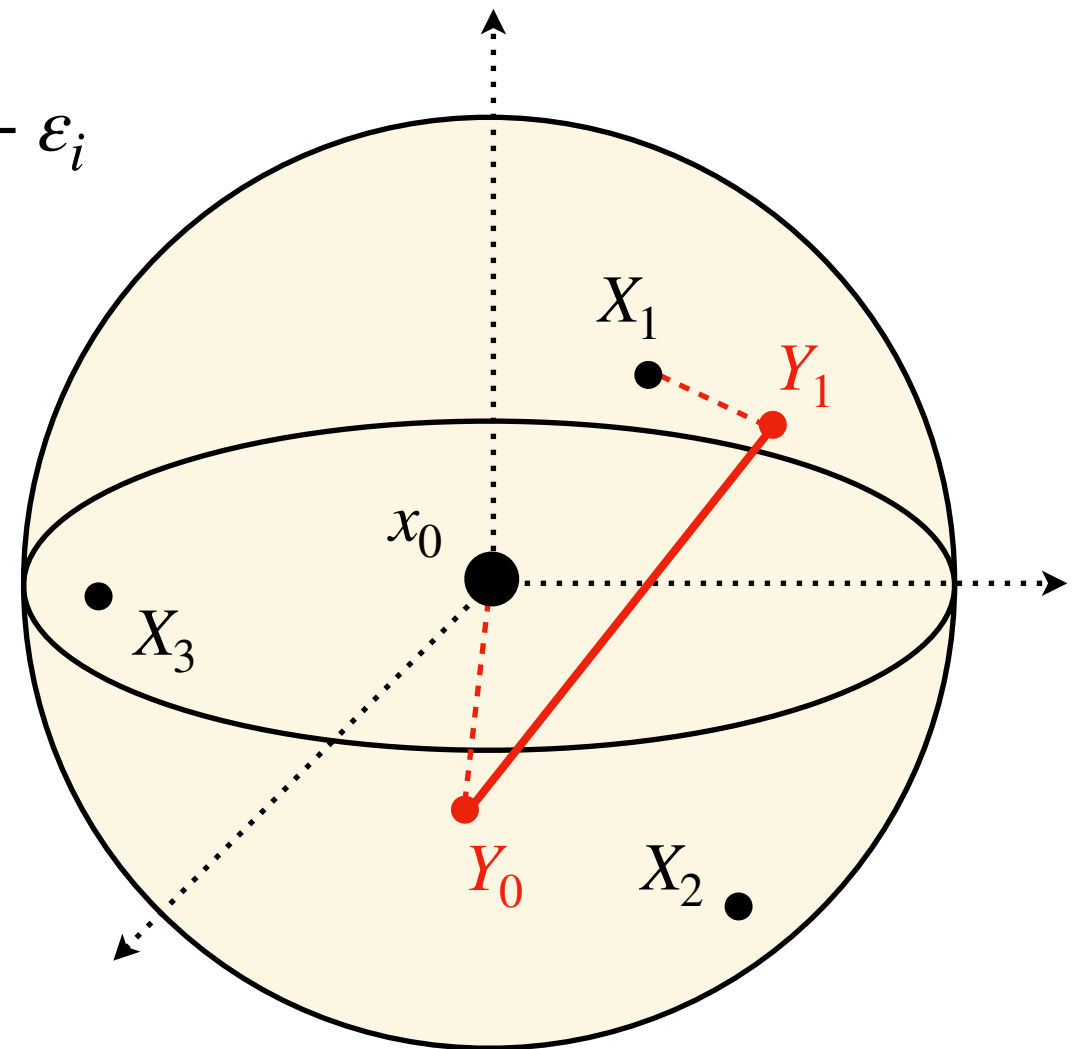
$$u_k^{IP}(m) \equiv - \mathbb{E} \left[\min_{1 \leq i \leq m} \|X_i\| \right]$$



AI-Mediated Search

Under **AI-mediated** search,

- The searcher's target is represented by $Y_0 = x_0 + \varepsilon_0$
- Each individual i is represented by $Y_i = X_i + \varepsilon_i$
- The noise terms $\varepsilon_0, \varepsilon_i \sim \mathcal{N}(0, \sigma^2 I_k)$ are independent of one another
- Henceforth refer to Y_i and Y_0 as *AI representations*

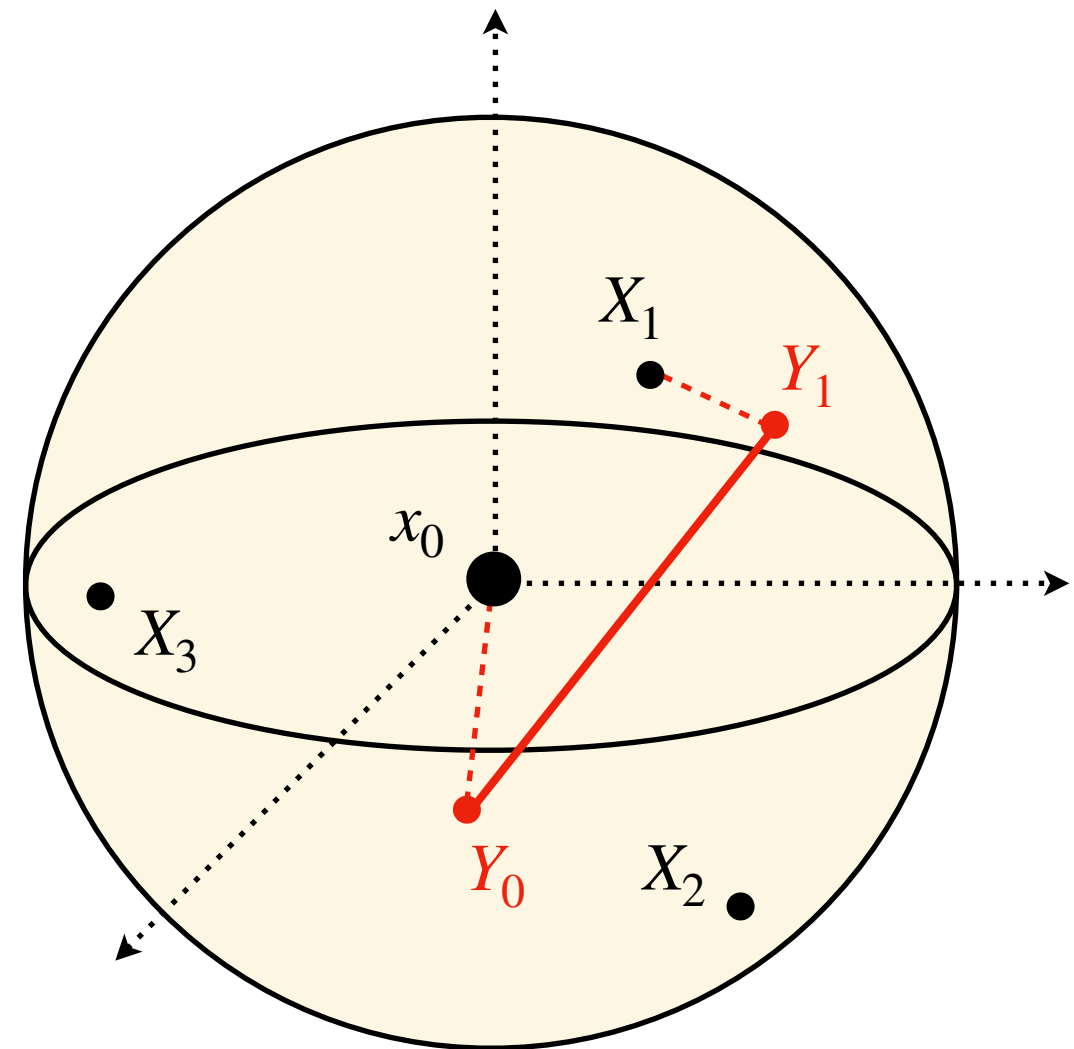


AI-Mediated Search

- The AI platform searches over n possible matches for the searcher
- The individual with the most compatible AI representation is

$$i_n^* = \operatorname{argmin}_{1 \leq i \leq n} \underbrace{\|Y_i - Y_0\|}_{\text{distance between the representation of the ideal point and the representation of individual } i}$$

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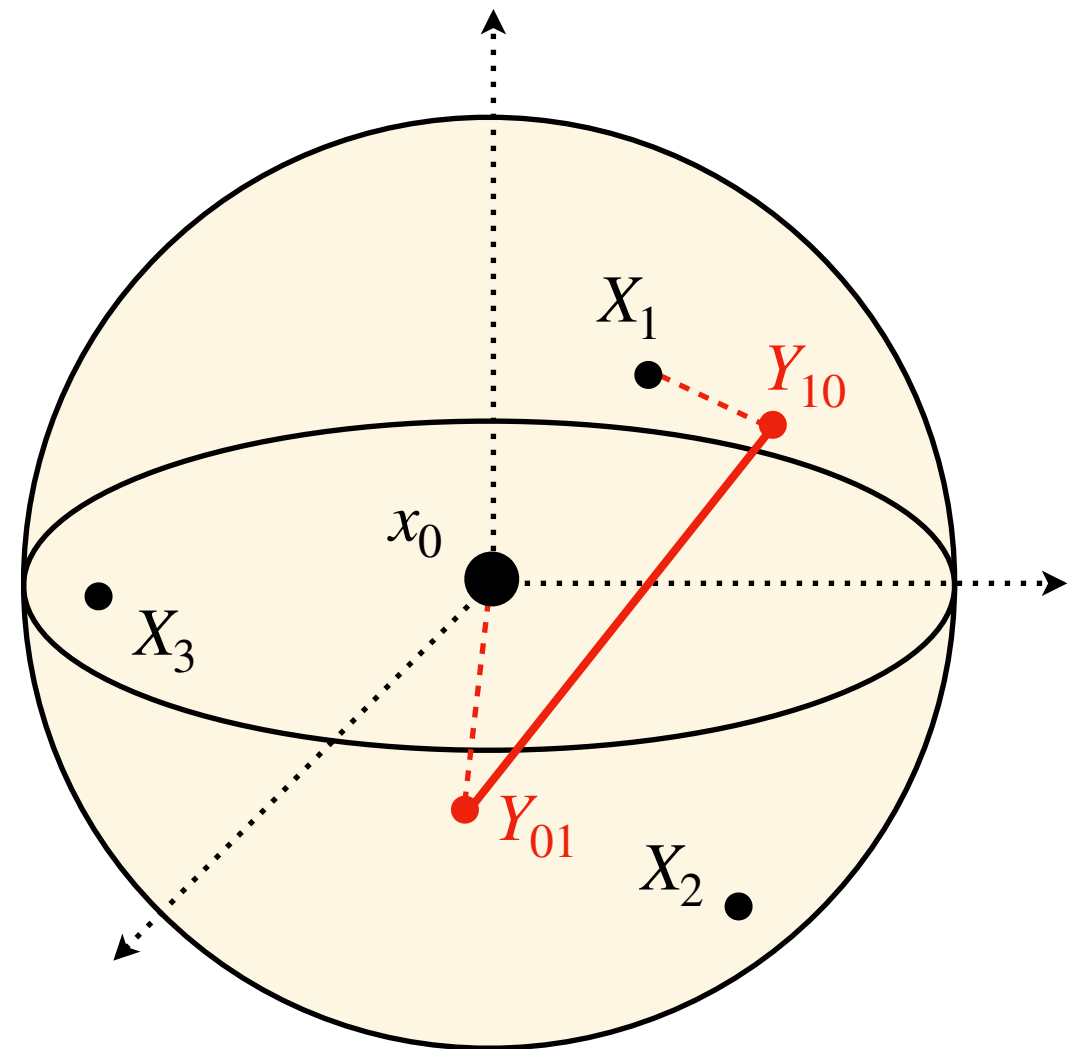
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- The searcher's expected utility is

$$u_k^{AI}(n) \equiv -E \left[\|X_{i_n^*}\| \right]$$

The selected match minimizes the distance between representations, but the searcher's payoff is based on the **true** distance between the underlying points



AI-Equivalent Sample Size

Measure of the value of AI-mediated search:

Definition: For any number of dimensions k , the *AI-equivalent sample size* m_k^* is the smallest $m \in \mathbb{Z}_+$ satisfying

$$u_k^{IP}(m) > u_k^{AI}(n) \quad \forall n \in \mathbb{Z}_+$$

if the inequality above is not satisfied for any finite m , then $m_k^* = \infty$.

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expected utility of best match
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in-person search over m people beats the AI platform, no matter how large its candidate pool

Interpretation of the Model

The model is motivated by AI clones but applies more generally to evaluation tasks that:

- (1) are **simple for humans** but difficult to **formally specify** for a machine
- (2) involve many attributes that are difficult to measure

Example: a family seeks to purchase a home

- in-person search: visit a small number of houses in person
- AI-mediated search: communicate preferences to an AI system, which (imperfectly) evaluates many homes with respect to this objective

The key comparison is between the capacity constraints of human evaluation and the errors of automated search

Main Results

Proposition: For every number of dimensions $k \in \mathbb{Z}_+$, the AI-equivalent sample size m_k^* is finite

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Theorem: For all k sufficiently large, the AI-equivalent sample size is $m_k^* = 2$

- When the number of relevant dimensions is sufficiently large, in-person search yields a better expected outcome so long as the searcher can consider at least two people
- Underscores the limits of AI representations for complex search problems

Proof Outline

expected utility:

expected distance:

in-person
search

$$u_k^{IP}(m) \equiv - \mathbb{E} \left[\min_{1 \leq i \leq m} \|X_i\| \right]$$

$$d_k^{IP}(m) \equiv \mathbb{E} \left[\min_{1 \leq i \leq m} \|X_i\| \right]$$

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need to show:

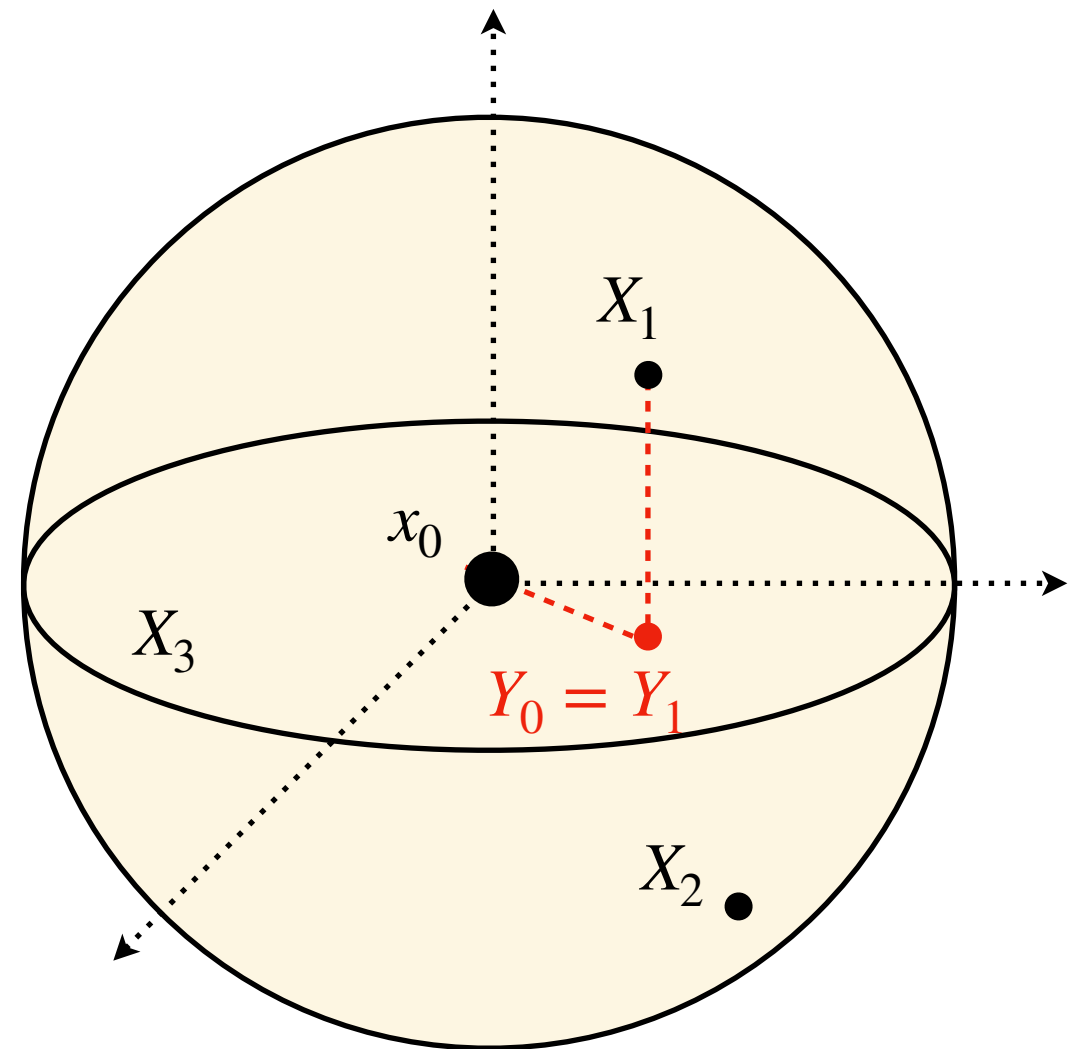
$$d_k^{IP}(2) < d_k^{AI}(n)$$

for all $n \in \mathbb{Z}_+$

Proof Outline

Define $d_k^{AI}(\infty) \equiv E(\|X_i\| : \|Y_i - Y_0\| = 0)$

- The expected distance to an individual whose AI representation perfectly matches the target representation
- Can interpret as completely saturated limit ($n = \infty$)



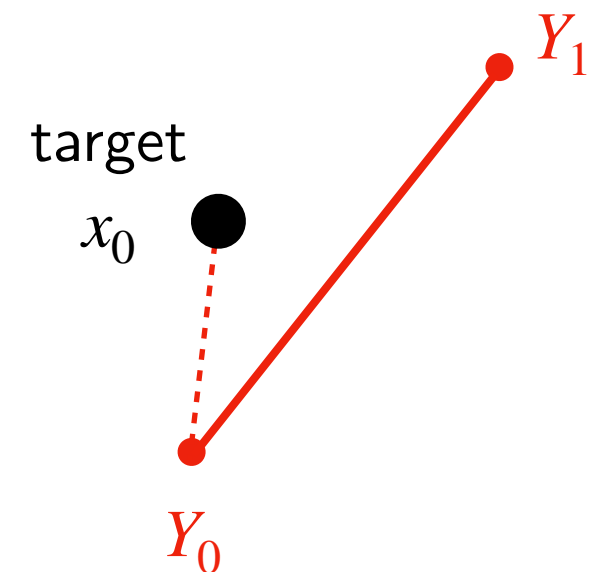
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Lemma: The pair $(\|X_i\|, \|Y_i - Y_0\|)$ has the monotone likelihood ratio property (MLRP)

- i.e., smaller distances between AI representations are associated with smaller actual distances
- nonstandard MLRP relation because of norms (not monotone)



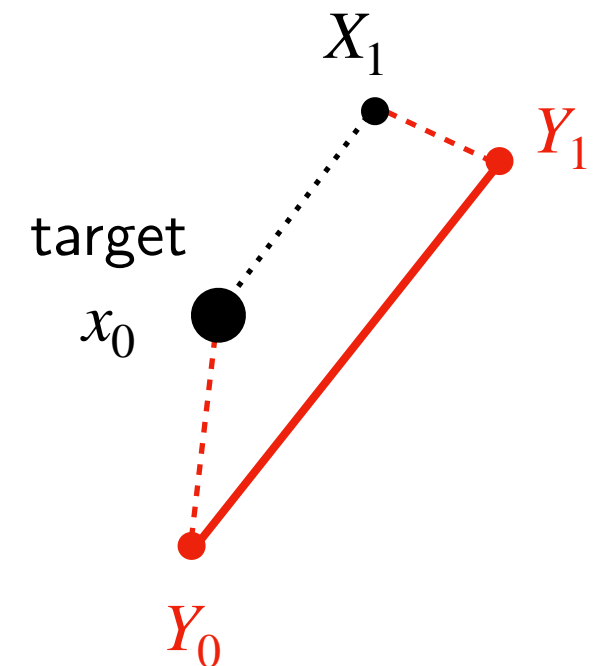
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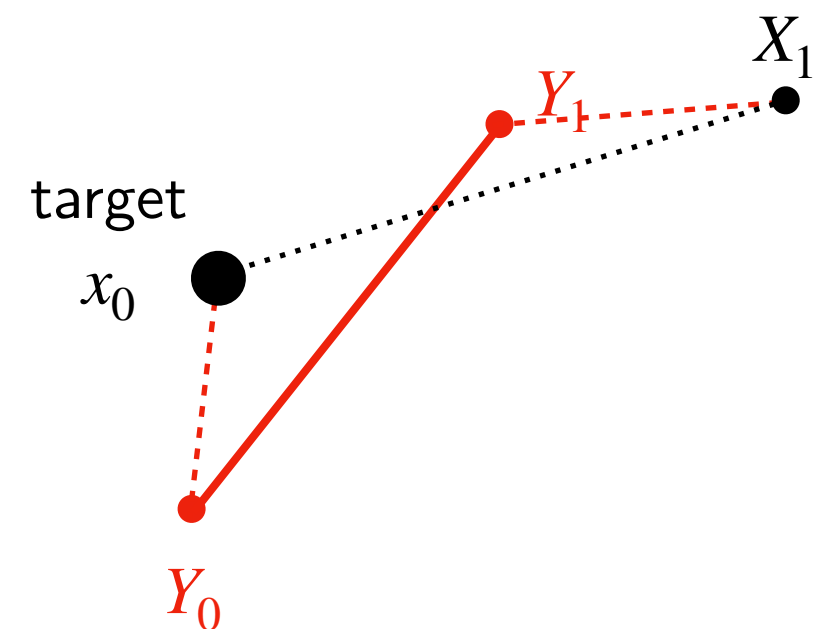
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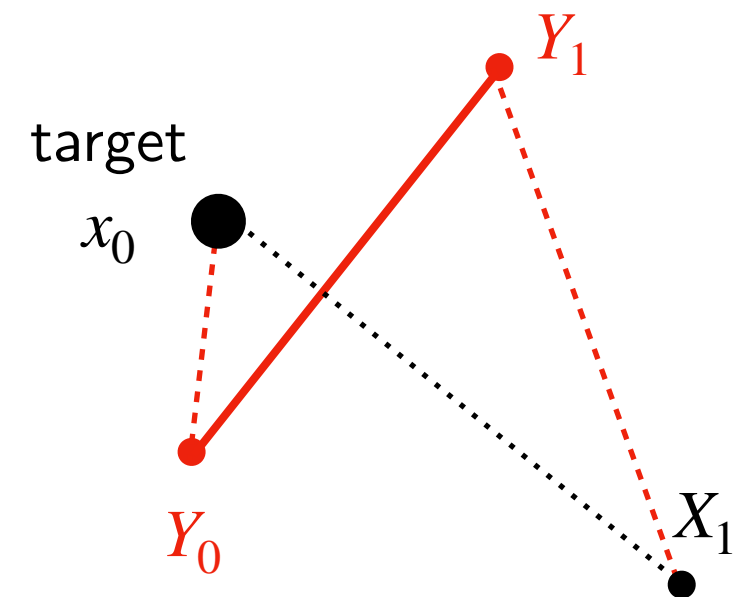
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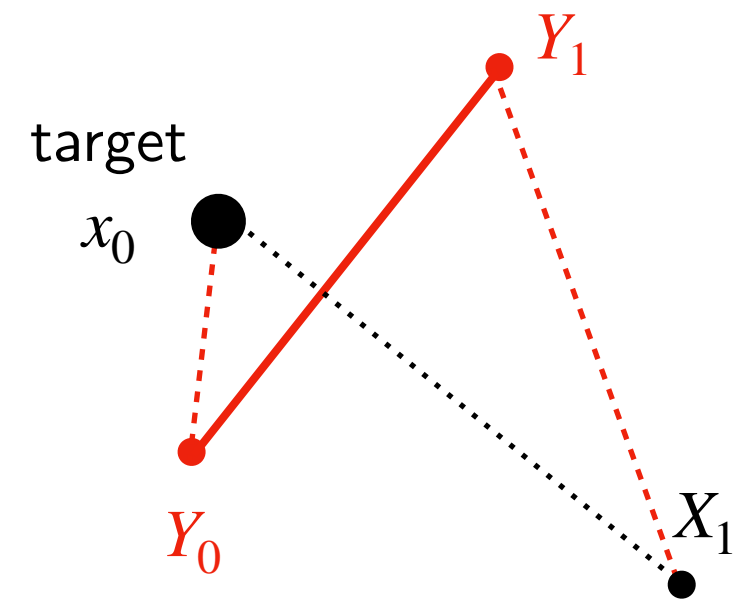
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Thus $E(\|X_i\| : \|Y_i - Y_0\| = s)$ is increasing in s

Thus $d_k^{AI}(\infty) \leq d_k^{AI}(n)$ for all n



Proof Outline

$$d_k^{IP}(2)$$

expected distance to best of 2 draws

$$d_k^{AI}(\infty)$$

expected distance to best match on AI
platform with infinite candidate pool

Proof Outline

$d_k^{IP}(2)$ expected distance to best of 2 draws

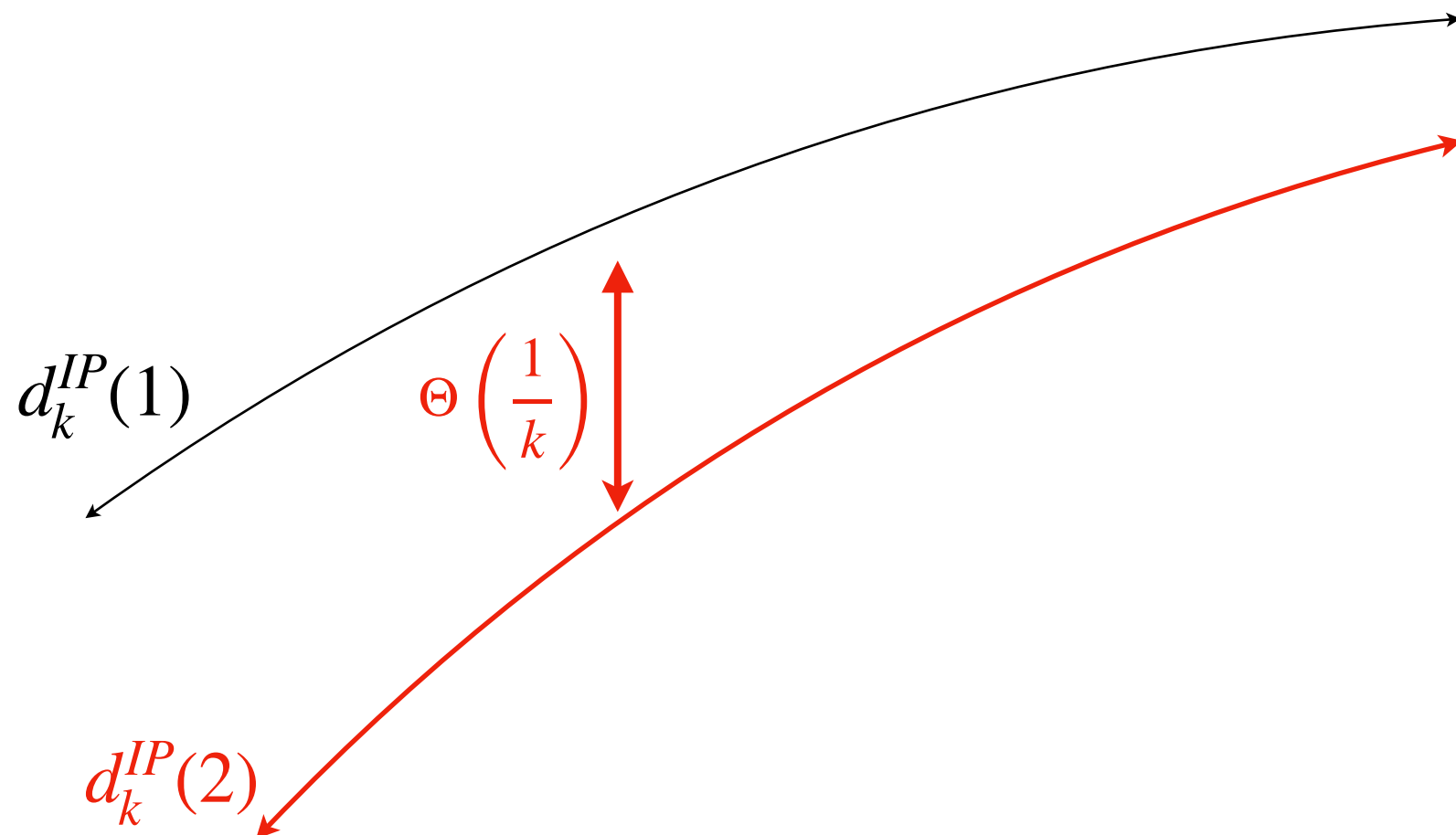
$d_k^{AI}(\infty)$ expected distance to best match on AI
platform with infinite candidate pool

benchmark:

$$d_k^{IP}(1) = \frac{k}{k+1} \quad \text{expected distance to 1 draw}$$

Proof Outline

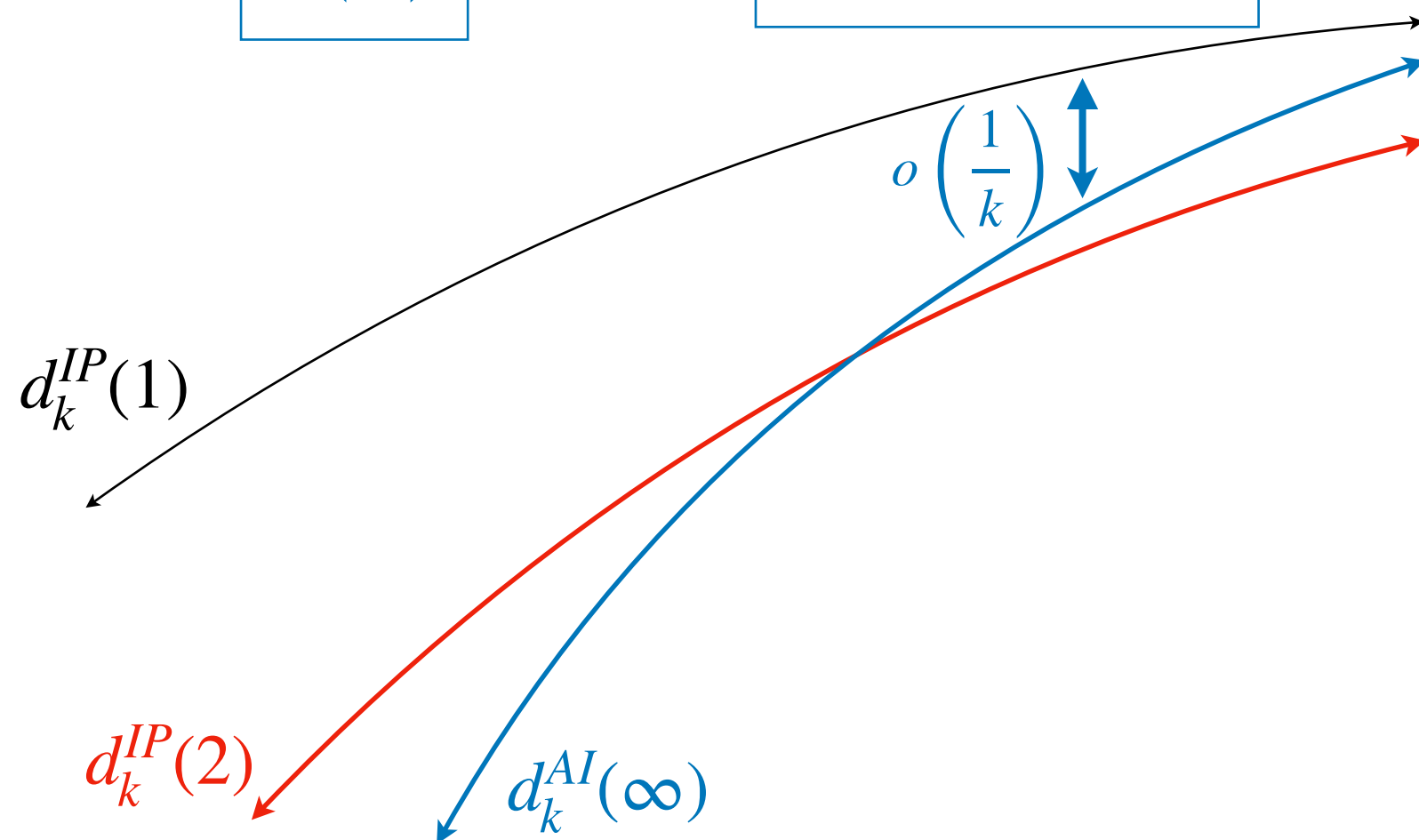
$$d_k^{IP}(1) - d_k^{IP}(2) = \Theta\left(\frac{1}{k}\right) \longrightarrow \text{vanishes at the rate } 1/k$$



Proof Outline

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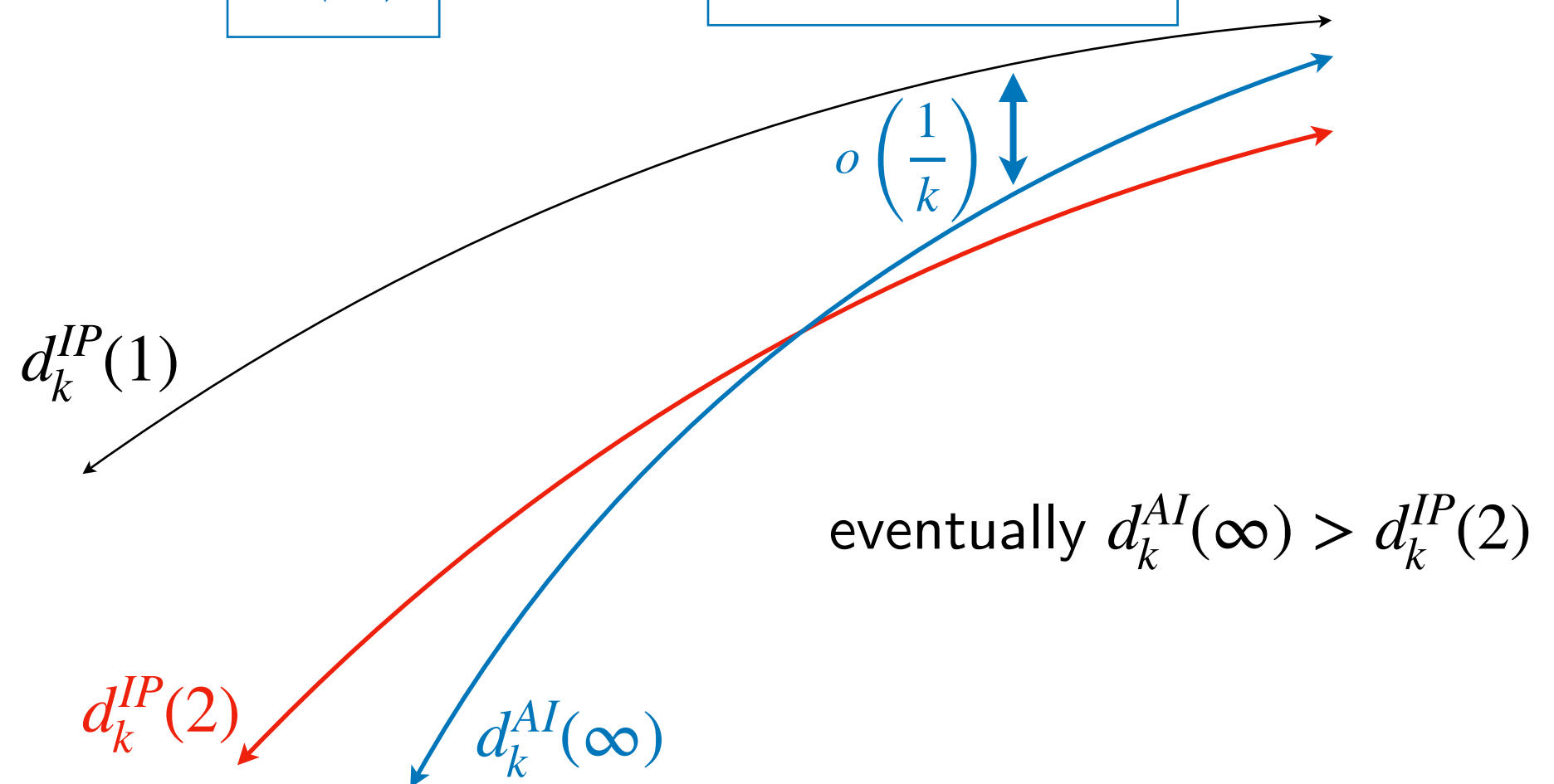
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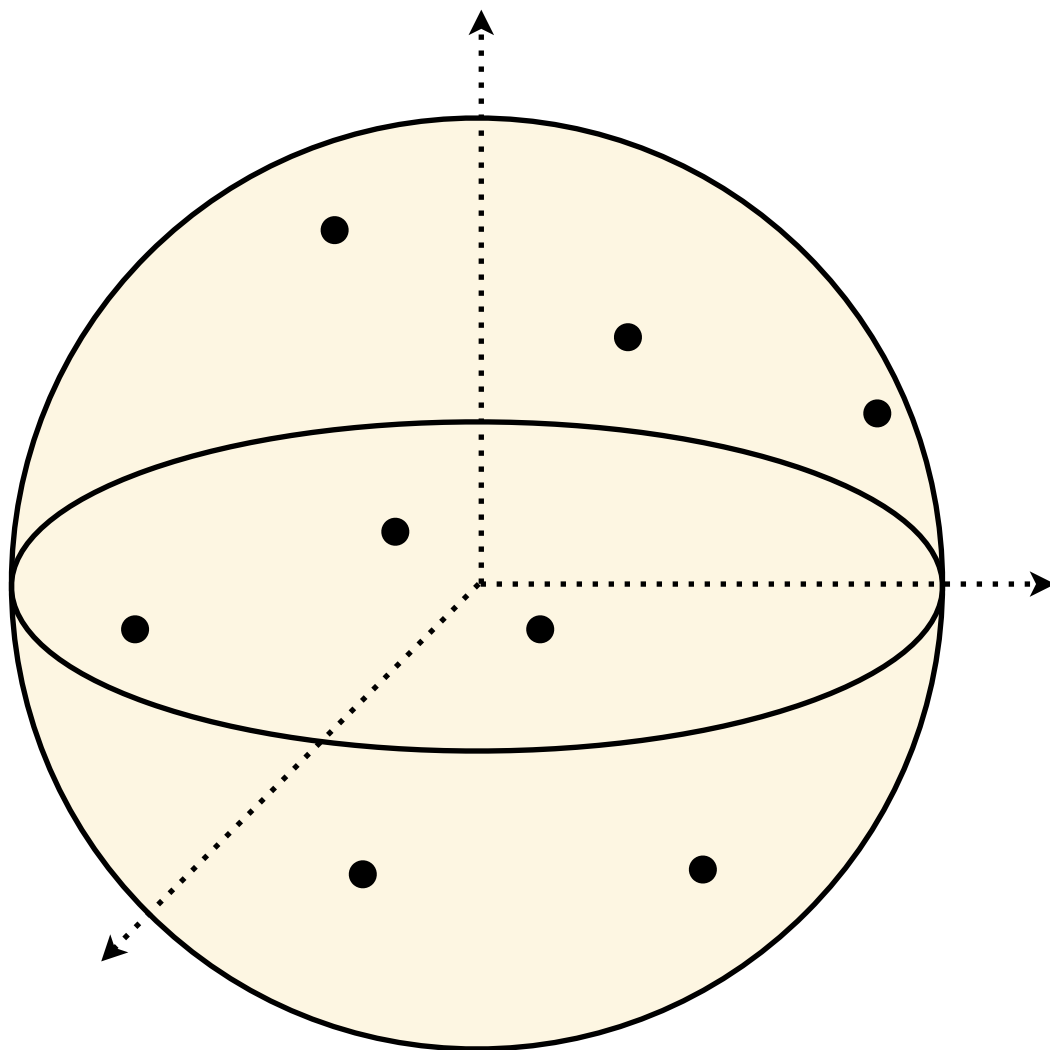
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Why does $d_k^{AI}(\infty)$ converge to $d_k^{IP}(1)$?

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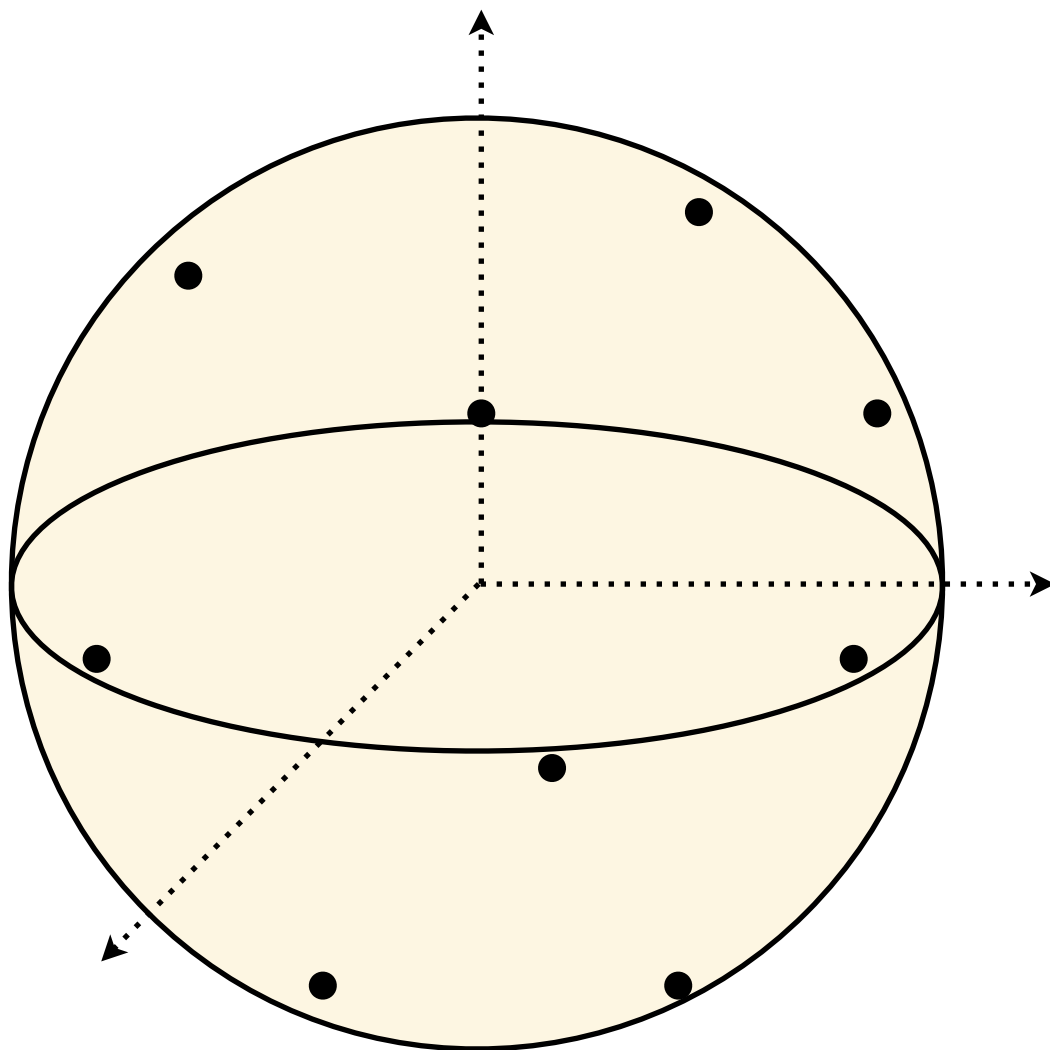
Might expect that points drawn uniformly at random from the unit ball look like this:



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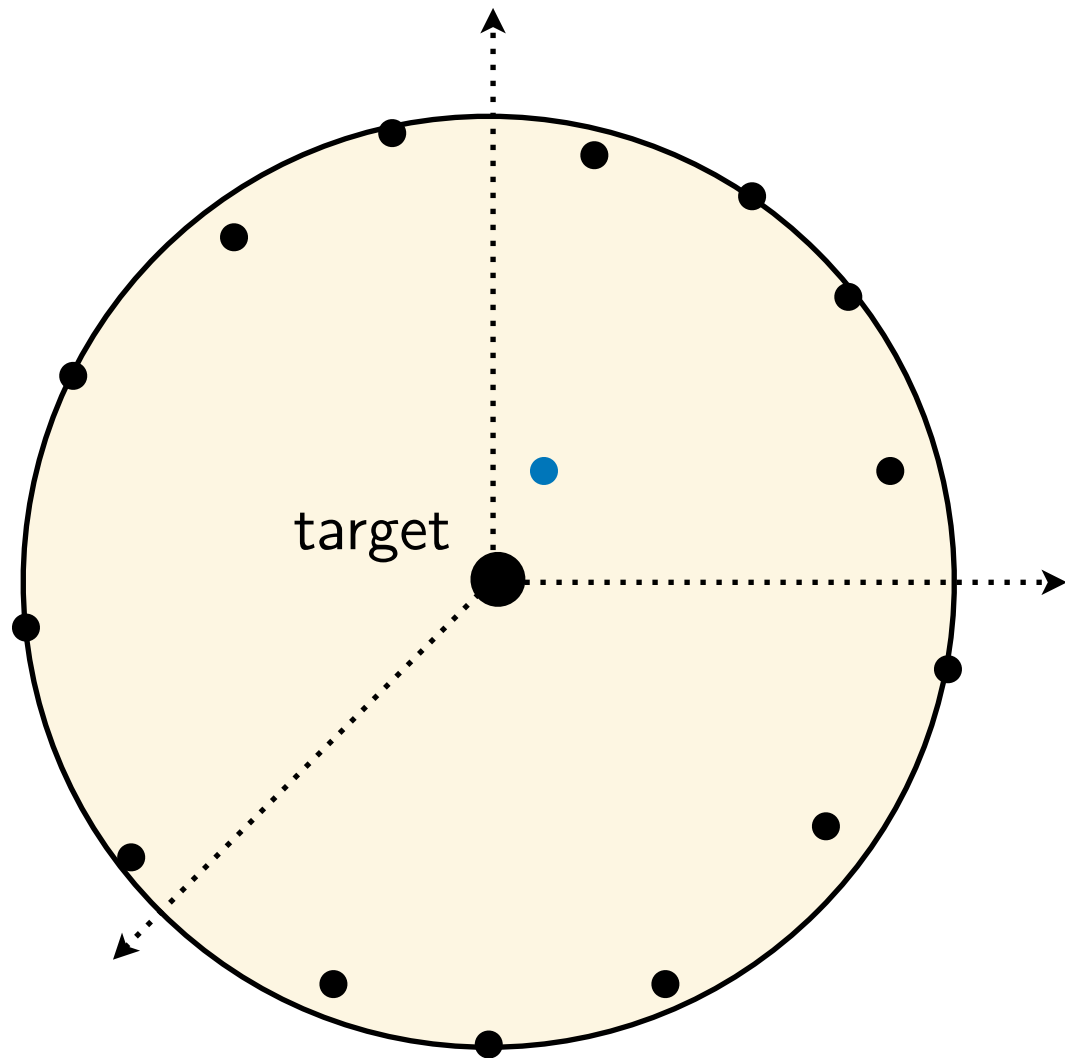
In fact, in high dimensions:

- points amass on the boundary of the unit ball
- distances between them concentrate around their expectations

i.e., people are increasingly unique and increasingly unlikely to exactly match the target

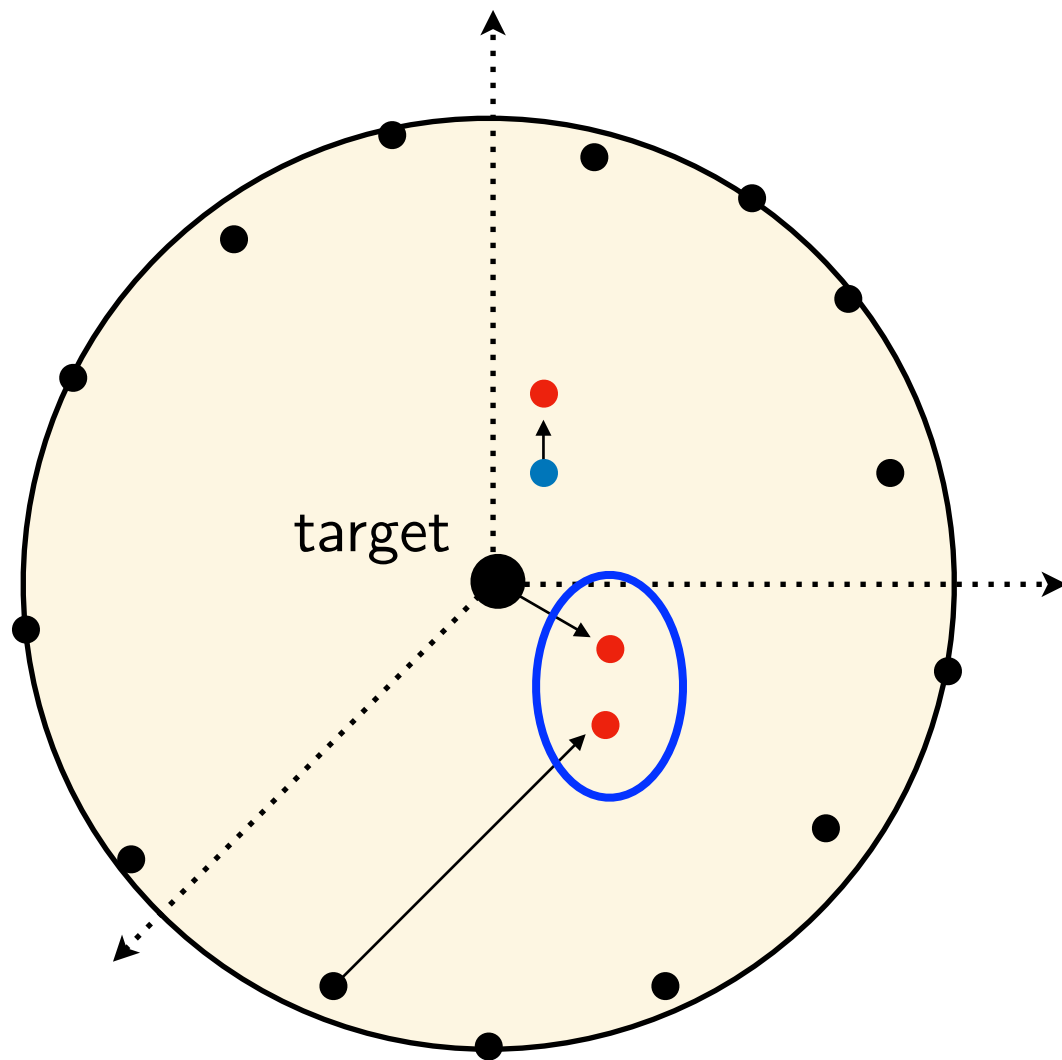
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Intuition

Why does $d_k^{AI}(\infty)$ converge to $d_k^{IP}(1)$?



- Best match increasingly driven by realization of the noise terms
- Eventually as if just selecting one candidate at random (i.e., $d_k^{IP}(1)$)

How Large is k ?

- In a simulation with $\sigma = 0.05$, the result holds for $k = 150$ dimensions

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- How many dimensions of personality are relevant for identifying
 - good spousal matches?
 - good employees?
- This is an empirical question

How Large is k ?

- In a simulation with $\sigma = 0.05$, the result holds for $k = 150$ dimensions
- How many dimensions of personality are relevant for identifying
 - good spousal matches?
 - good employees?
- This is an empirical question
- The main takeaway is that **the complexity of match compatibility matters** for how we should think about human vs. AI search
- The value of AI representations can differ across search problems depending on what the relevant k is

Extended Model

- In practice there is heterogeneity in the AI's ability to represent different individuals
- Many reasons for this
 - different quantities of personal data to share with AI (e.g., because of variation in social media usage)
 - different willingness to share it (e.g., because of variation in privacy preferences)
- Next suppose the population is split into a “data-rich” and “data-poor” group

Extended Model

- Let $g \in \{R, P\}$ denote the group (data-rich or data-poor)
- Groups are identical in characteristics: each X_i^g independently drawn from a uniform distribution on the unit ball
- Individuals i from group g have AI representations

$$Y_i^g = X_i^g + \varepsilon_i^g, \quad \varepsilon_i^g \sim \mathcal{N}(0, \sigma_g^2 I_k)$$

where $\sigma_R^2 < \sigma_P^2$ (lower error for data-rich group)

- The AI platform searches over n candidates from each group

Extended Model

Proposition: in the limit as the population grows large, the probability that the searcher's match is from the data-rich group **exceeds $1/2$**

- Although the two groups have identical compatibility with the searcher, a data-rich individual is more likely to be identified as the best match

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Proposition: in the limit as the number of dimensions and population size both grow large, the probability that the searcher's match is from the data-rich group **converges to 1**

Implications

- This systematic selection of data-rich individuals is not harmful to Subject:
 - The true distance between Subject and the selected individual is (in expectation) smaller if the individual is data-rich
 - Their AI clones more accurately represent them
- But this selection process has potentially concerning distributional implications
 - Individuals who are better understood by AI systems will be systematically preferred
 - This advantage could exacerbate existing inequalities
 - Moreover points to an emerging form of social stratification: an individual's opportunities may depend not only on their qualities, but also on how effectively AI systems can capture those qualities

Creative Ownership in the Age of AI

Annie Liang
(Northwestern)

Jay Lu
(UCLA)

Introduction

Have already discussed how generative AI systems can now create images, novels, persons, etc.

These capabilities depend crucially on having been trained on a large body of human-authored work.

AI creations can closely imitate the style of those original works without copying content, a capability now central to ongoing litigation.

- Andersen v. Stability AI
 - Disney Enterprises v. Midjourney
 - Universal Music Group Recordings v. Suno
- + many others

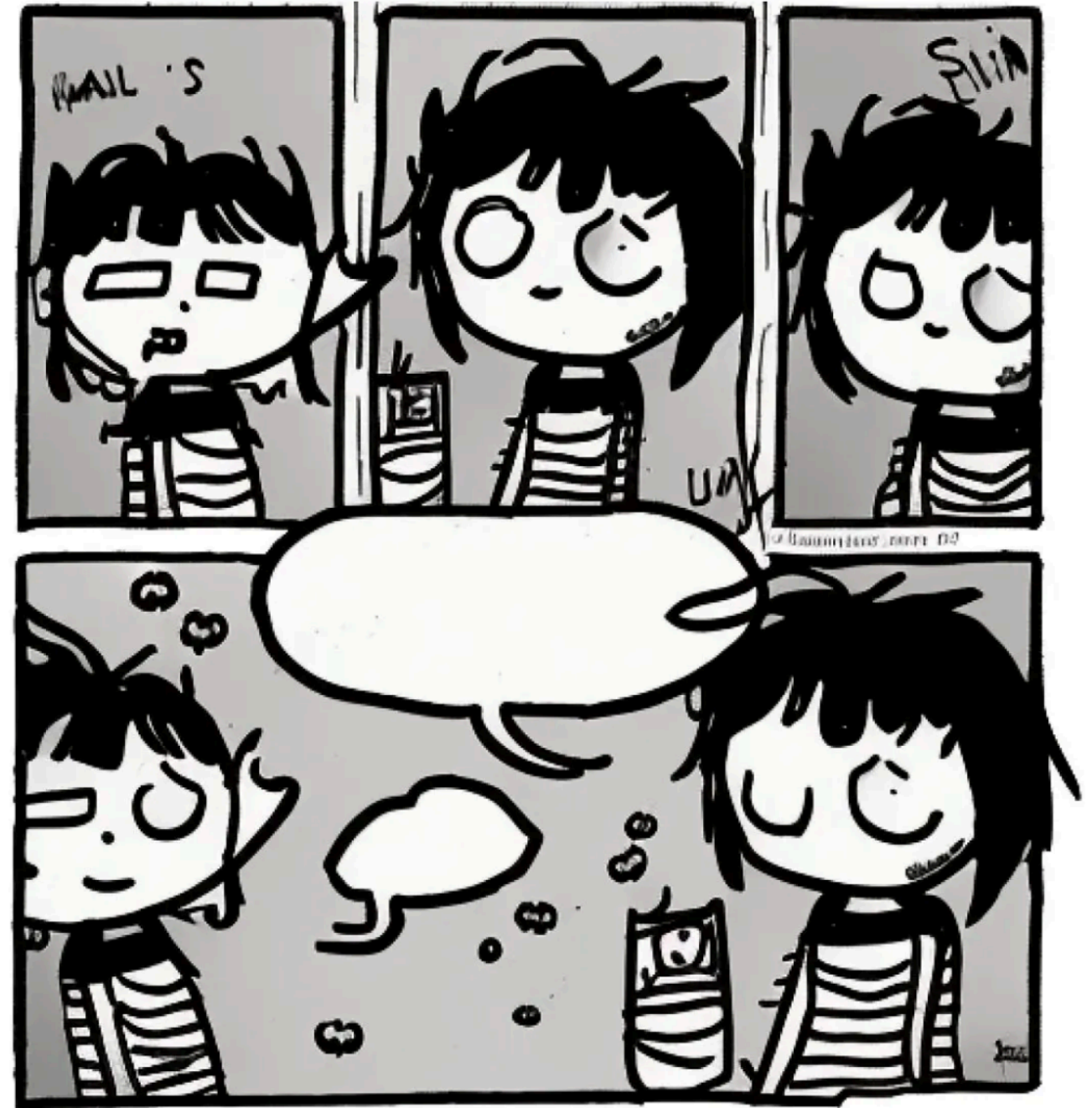
Introduction

A recent example of my work



Sarah Andersen, *NYTimes*, 2022

An A.I. generated image that I created when I used my name as a prompt



It's not perfect — but it has captured the signature elements of my drawing style. The black bangs, striped shirt and wide eyes are immediately recognizable. As A.I. gets increasingly sophisticated, I am growing more concerned about what's possible.

Legal Context

Existing copyright doctrine applies to GenAI at two stages:

1. **Training** on copyrighted works may violate copyright law.

- Possibly too strong: human authors learn by reading copyrighted books.
- May overly constrain AI capabilities to not allow it to learn in a similar way
- Moreover, fair use doctrine suggests that training on copyrighted works may be OK if the use is sufficiently “transformative”

2. **Generated outputs** may violate copyright law if they are “substantially similar” to an existing work.

- Copyright law has historically drawn a line between protectable **expression** and unprotectable **style** to preserve creative competition and follow-on innovation.
- But did not account for the emergence of AI systems that could easily, flexibly, quickly mimic style

This Paper

Existing notions of copyright and intellectual property **may not be appropriate for a post-generative AI world.**

This paper:

- We propose a new criterion for evaluating when an output is derivative of an existing work:

Would the output have been generable without that existing work in its training corpus?

- Study the implications of this criterion for permissible generation
- Whether the creative process is **light or heavy tailed** matters critically for the restrictiveness of this criterion

Model

Setup

- A *creation* is a vector $x \in \mathbb{R}^d$
- A *corpus* is any (Borel measurable) subset of creations $C \subset \mathbb{R}^d$
- Let $\mathcal{C}$ denote the set of all possible corpora
- A *generator* is a technology for producing new creations from an existing corpus.

Generators

Definition: The map $g : \mathcal{C} \rightarrow \mathcal{C}$ is a *generator* if for all $C, C' \in \mathcal{C}$ it satisfies:

1. Preservation: $C \subseteq g(C)$ — *reproduces all input creations*
2. Monotonicity: $C \subseteq C'$ implies $g(C) \subseteq g(C')$ — *weakly expands the set of generable creations when the original corpus expands*
3. Idempotence: $g(g(C)) = g(C)$ — *running it on its output produces no new creations*

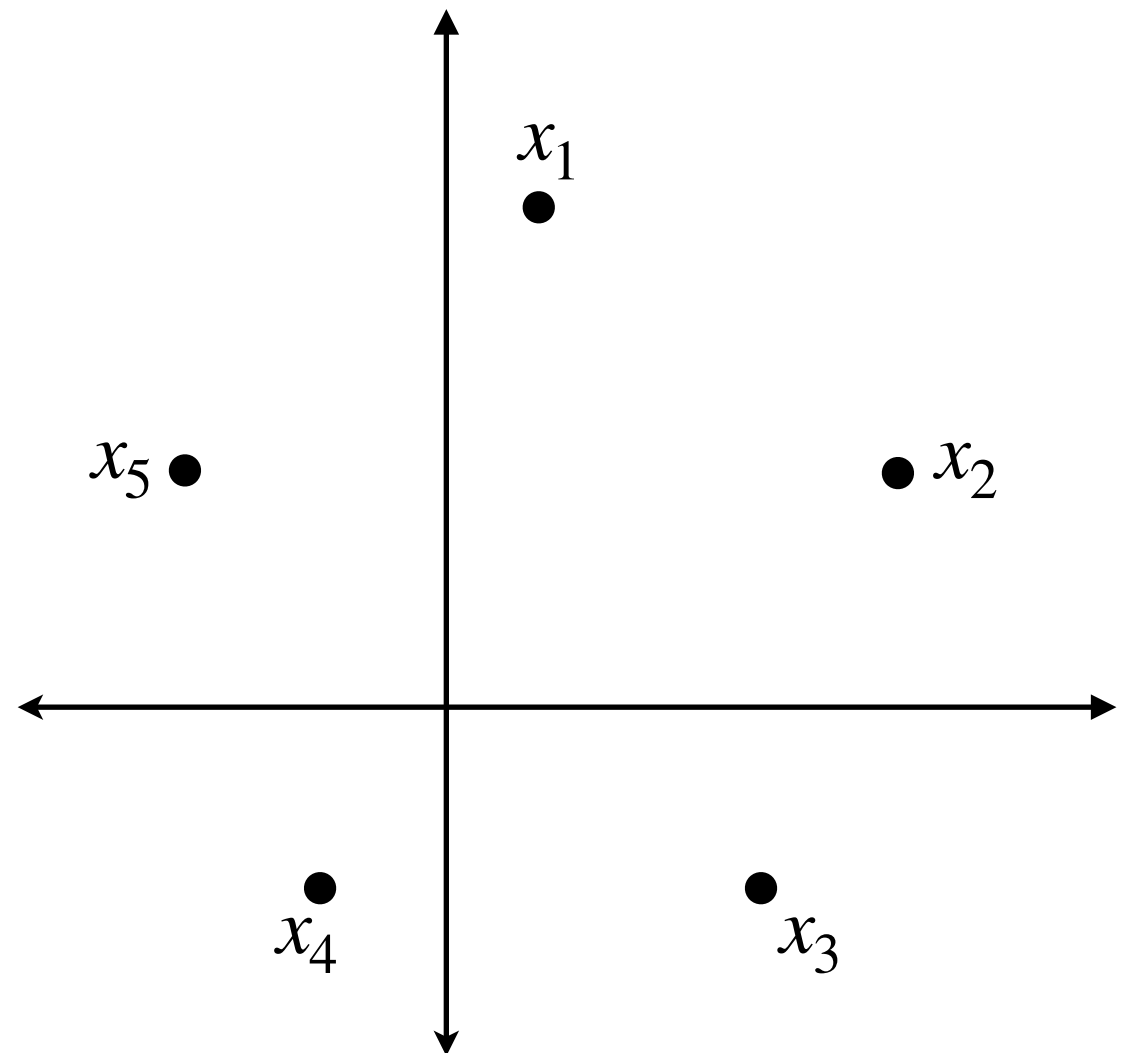
These are also known as closure operators.

We interpret $g(C)$ interpreted as the set of all possible creations given the initial training corpus C .

Example Generators

Convex Hull Generator: $g_{conv}(C) = \text{Conv}(C)$

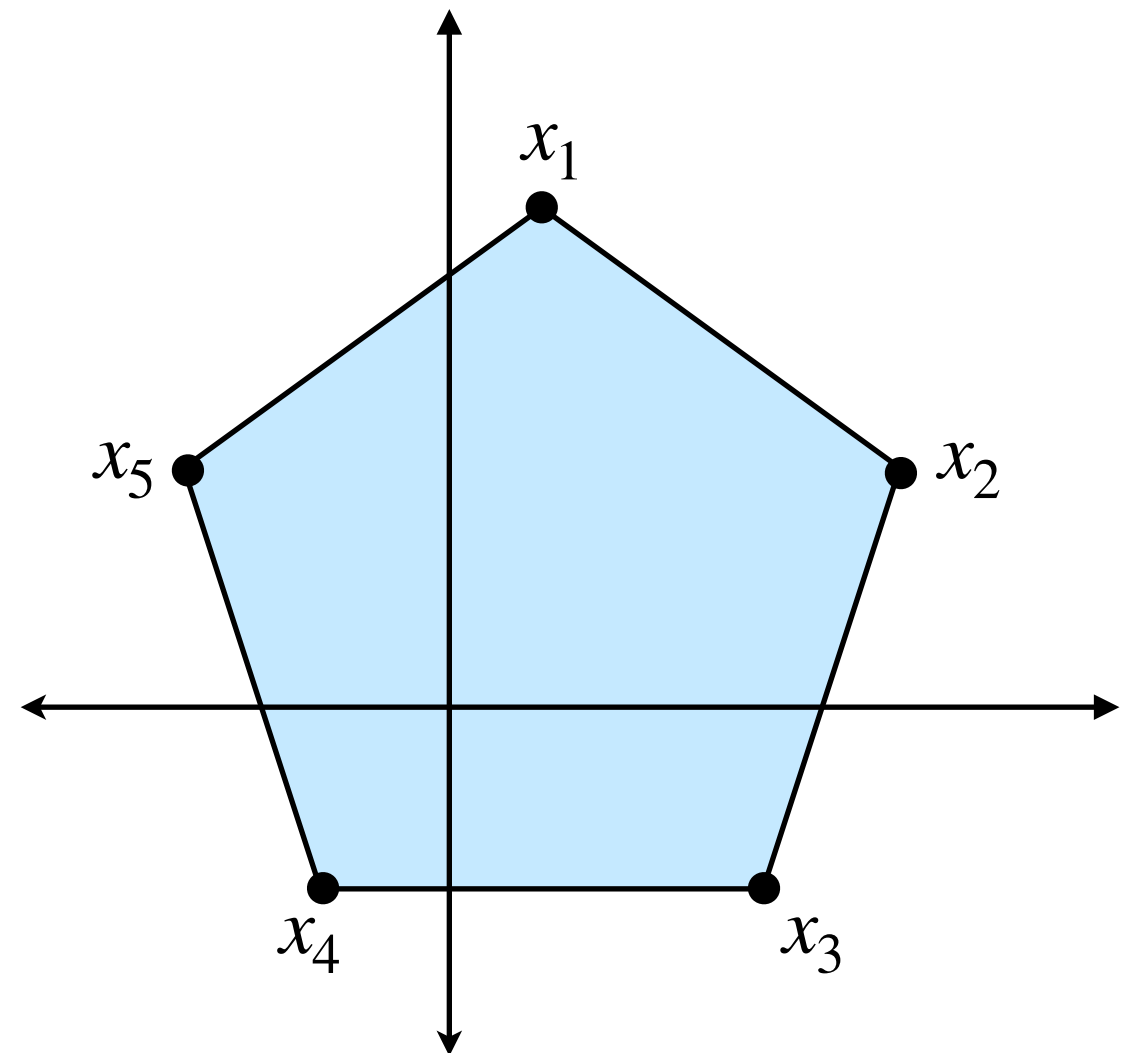
- Generate any convex combination of existing creations



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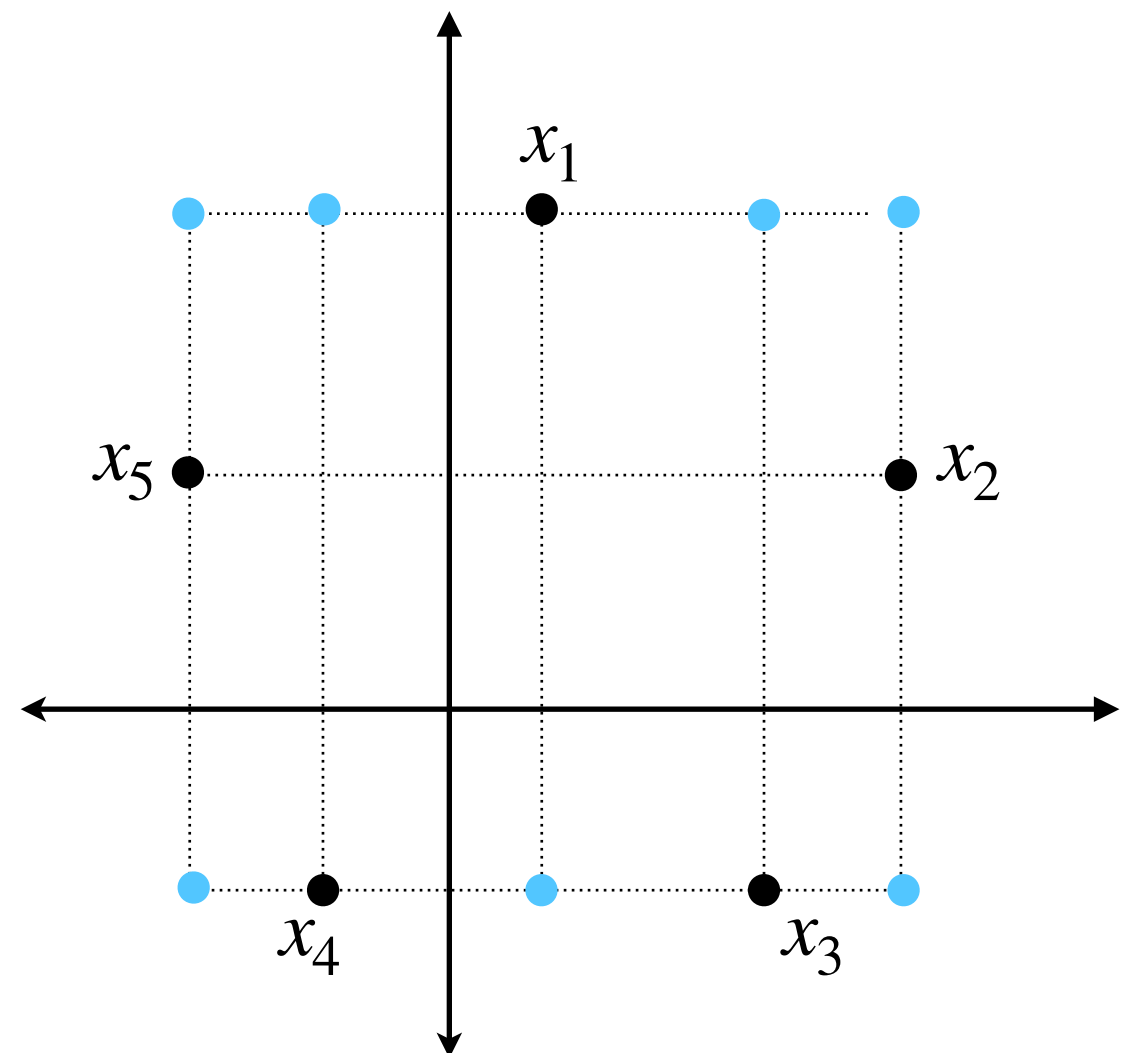
Example Generators

Convex Hull Generator: $g_{conv}(C) = \text{Conv}(C)$

- Generate any convex combination of existing creations

Splice Generator: $g_{splice}(C) := \{x \in \mathbb{R}^d : \forall k = 1, \dots, d, \text{ there is a } x' \in C \text{ s.t. } x_k = x'_k\}$.

- Recombines features of existing creations



Example Generators

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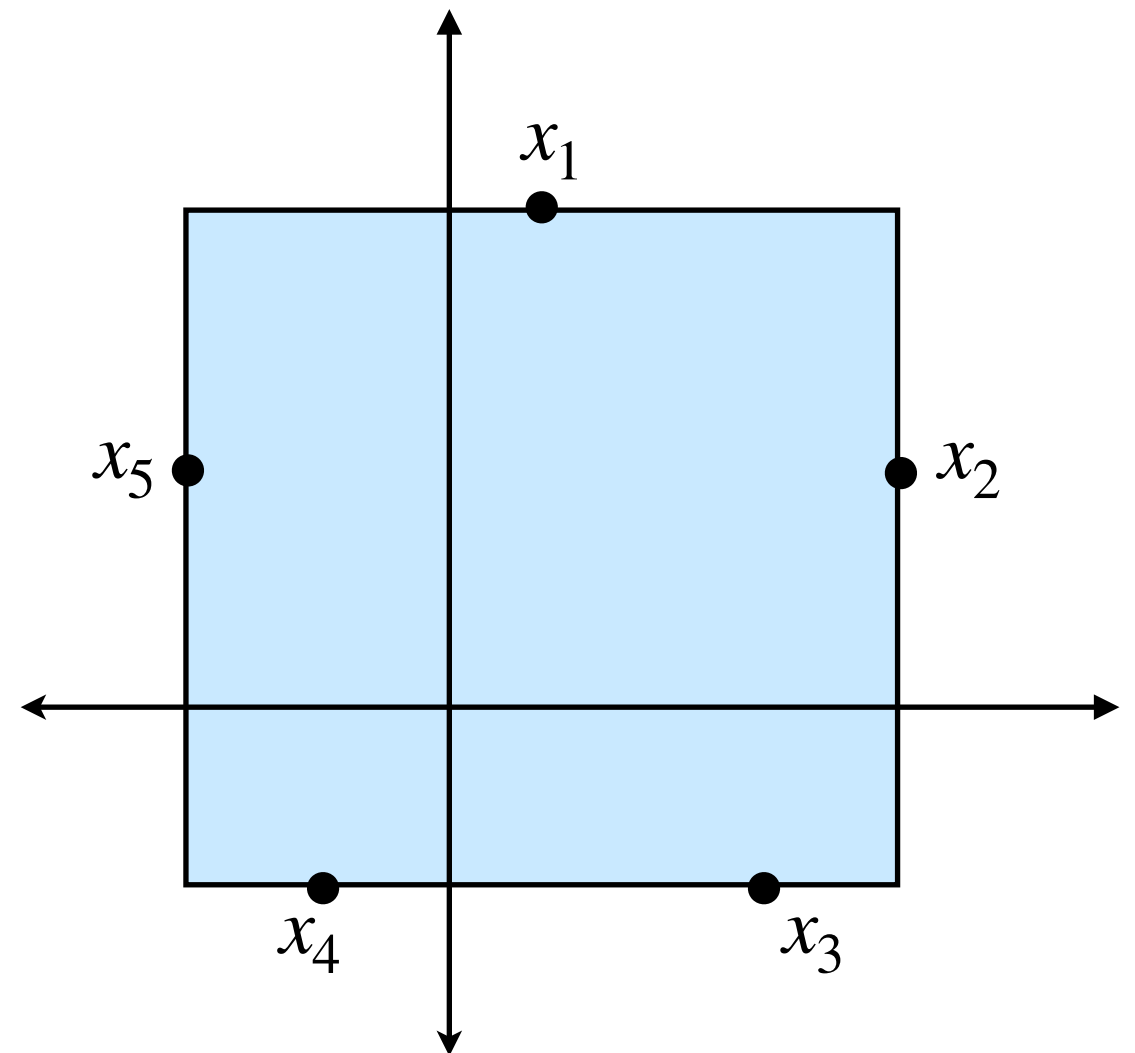
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- Recombines features of existing creations

Box Generator: $g_{box}(C) = g_{conv}(g_{splice}(C))$

- Composes the convex hull and splice generator



Example Generators

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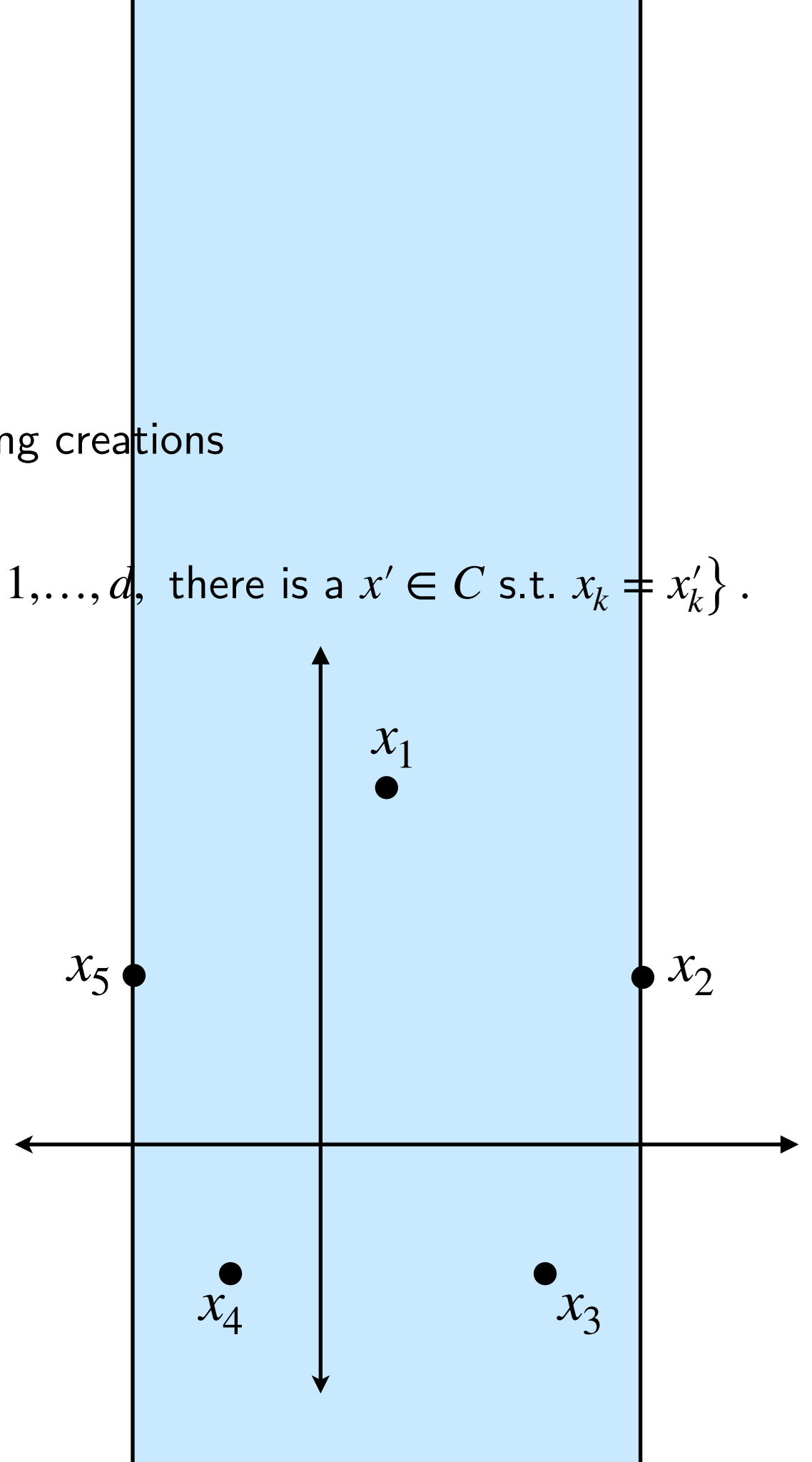
- Recombines features of existing creations

Box Generator: $g_{box}(C) = g_{conv}(g_{splice}(C))$

- Composes the convex hull and splice generator

v -Transform Generator:

- Shift the convex hull along a fixed direction v .



Convex-Valued Generators

Definition: A generator g is *convex-valued* if $g(C)$ is convex for every $C \in \mathcal{C}$.

- Includes the convex-hull generator, box generator, and the v -transform generator (but not the splice generator)

Result: The following are equivalent:

1. g is convex-valued
2. $g = g_{conv} \circ g$
3. $g = g \circ g_{conv}$

- Thus can interpret restriction to convex-valued generators as an assumption that the convex-hull generator is available
- Some of our results will make this restriction

Permissible Generation

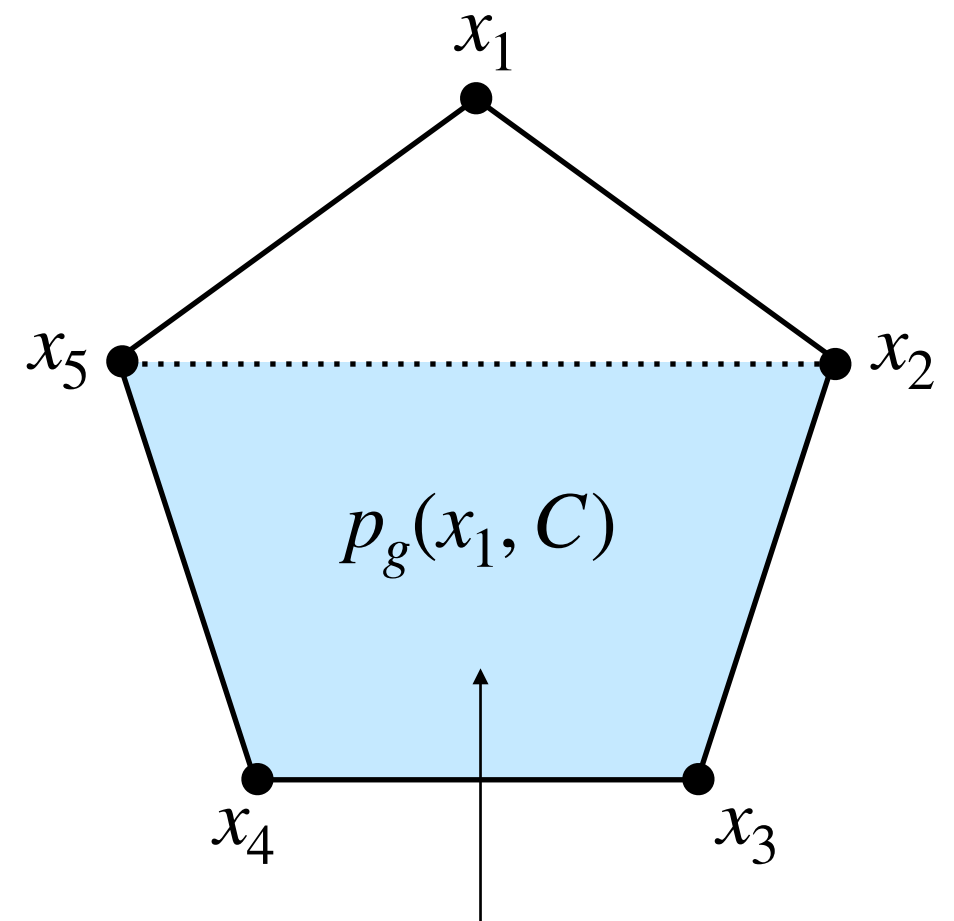
- Our proposed criterion specifies when a generated creation critically depends on the presence of some existing creation.
- Would the creation still be generable without that existing work in the corpus?

Permissible Generation

Definition: Fix a generator g and corpus C . For each existing creation $x \in C$:

1. The x -**permissible set** is $p_g(x, C) := g(C \setminus \{x\})$, i.e., all creations that are generable from $C \setminus \{x\}$
2. The x -**violation set** is $v_g(x, C) := g(C) - p_g(x, C)$, i.e., all creations that are generable from C but not from $C \setminus \{x\}$

Convex hull generator:



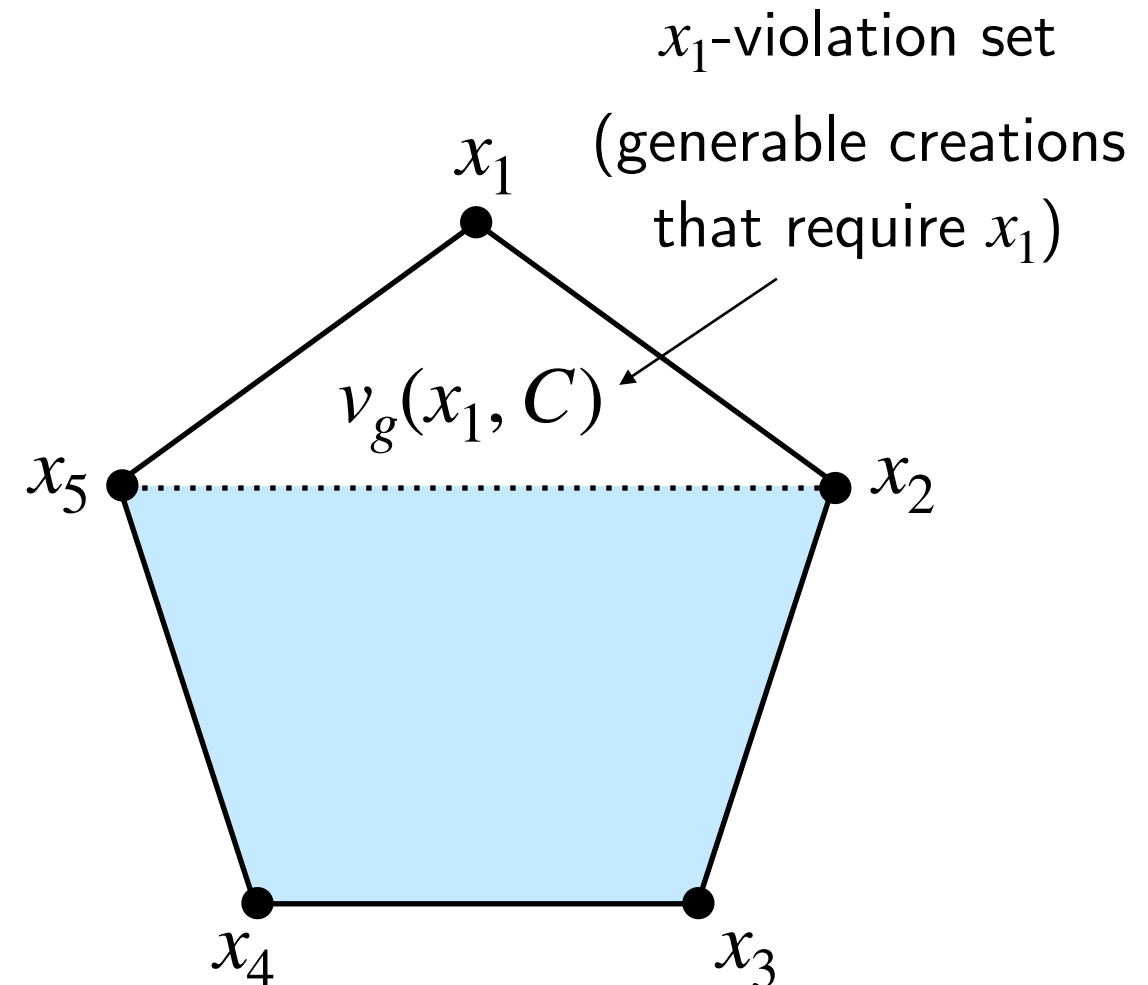
x_1 -permissible set
(generable creations that do not require x_1)

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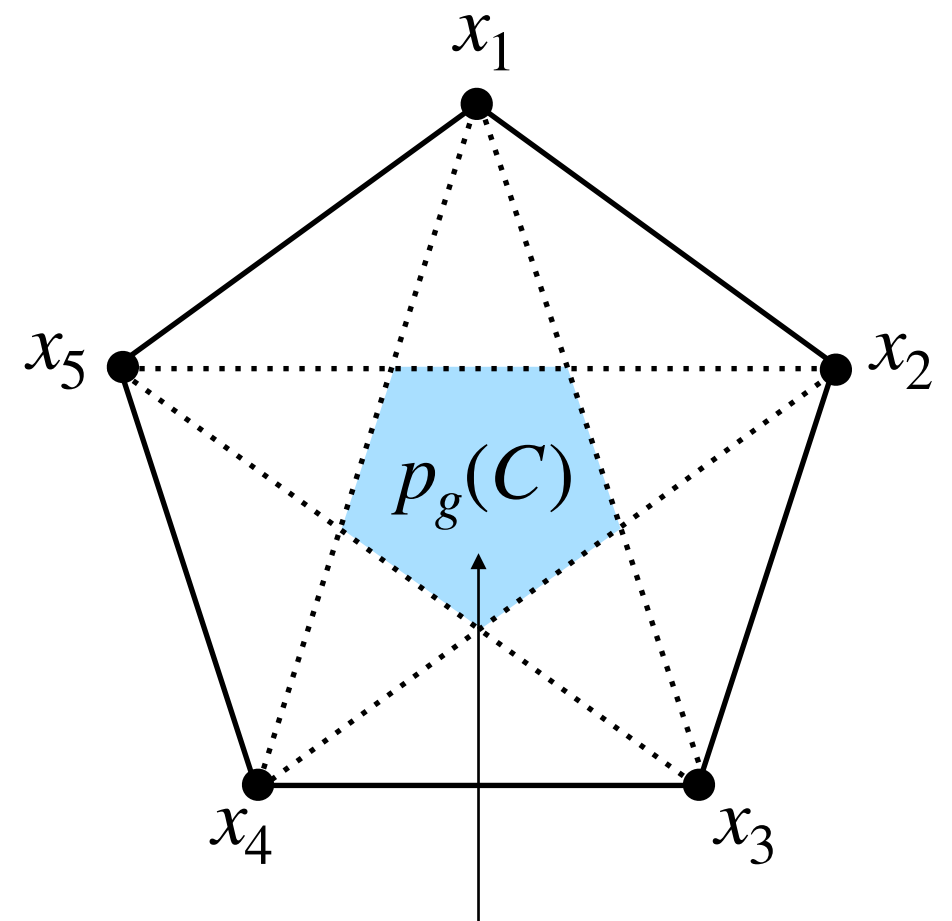
The **permissible set** is

$$p_g(C) := \bigcap_{x \in C} p_g(x, C)$$

The **violation set** is

$$v_g(C) := \bigcup_{x \in C} v_g(x, C)$$

Convex hull generator:



permissible set

(generable creations that do not critically rely on any $x \in C$)

Permissible Generation

Definition:

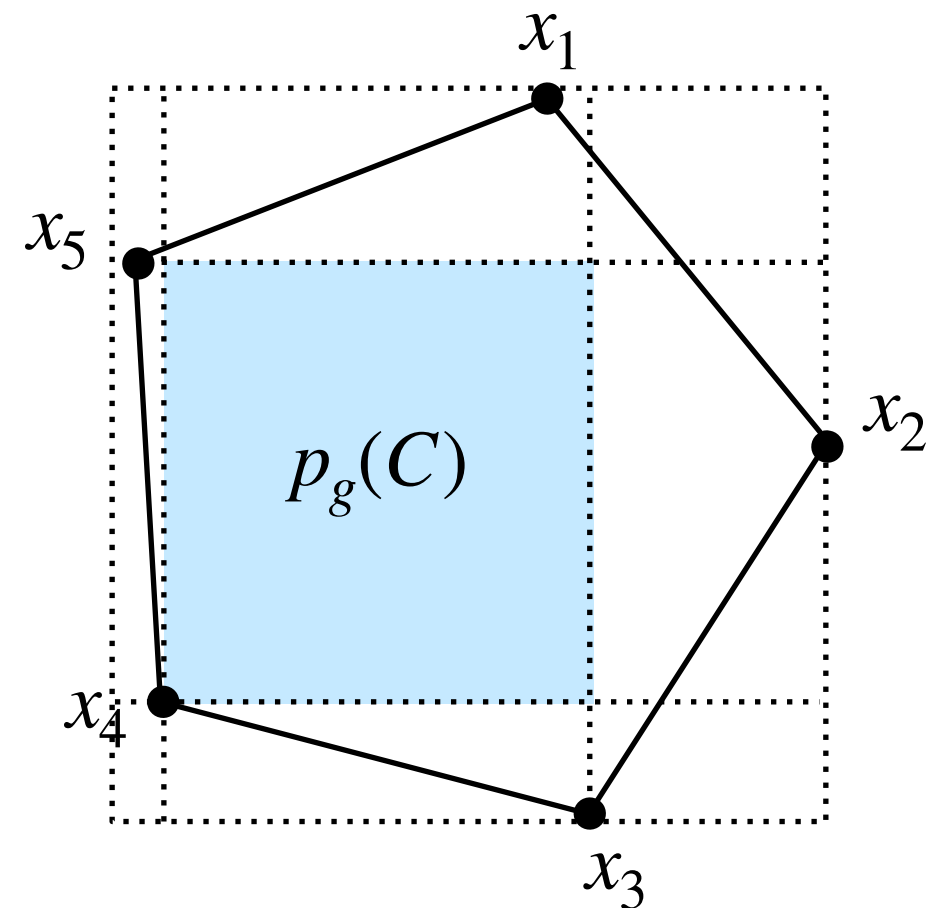
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Box generator:



Note: Boundary points **can** be permissible, depending on the generator and corpus

Interpretation

- If a generated creation lies in the **violation set**, then **at least one existing work** has a claim under our criterion.
- If a generated creation lies in the **permissible set**, then **no existing work** has such a claim.
- What does it mean for an existing work to “have a claim”?
 - The predecessor may be entitled to **attribution**.
 - The predecessor may be entitled to **financial compensation or rent**.
 - The downstream work may be **restricted or prohibited**.
- Our framework is agnostic about which, our goal is to understand which/how many generated outputs are permissible under the proposed criterion.

In the Paper

Having set up this framework, there are many questions one could ask about generation and permissible generation.

The paper establishes several properties of the permissible set that I will skip today.

Focus instead on how restrictive our criterion is, and how that evolves with the corpus itself:

- When the corpus is large, does infringement become a vanishing concern, or does it remain a persistent feature of generative technology?
- The answer depends on our assumptions about the process of creative generation, and the rarity of breakthrough innovations.

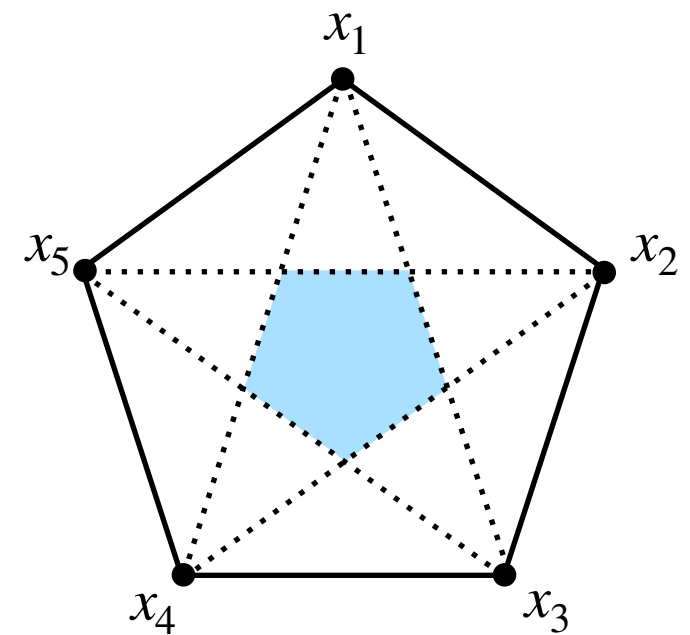
Properties of Permissible Creation

Permissible Ratio

Let $\text{Vol}(A)$ denote the Lebesgue measure (or *volume*) of the set $A \in \mathcal{C}$.

Definition: The **permissible ratio** is:

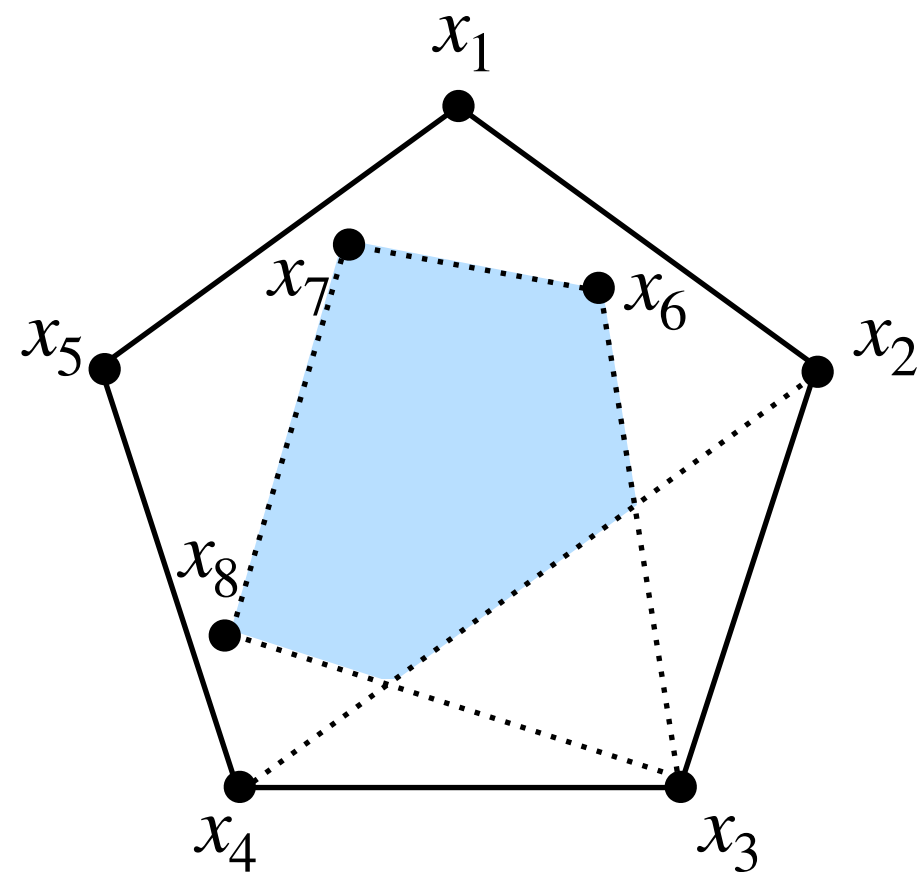
$$r_g(C) := \frac{\text{Vol}(p_g(C))}{\text{Vol}(g(C))}$$



i.e., the proportion (by volume) of generable creations that are permissible.

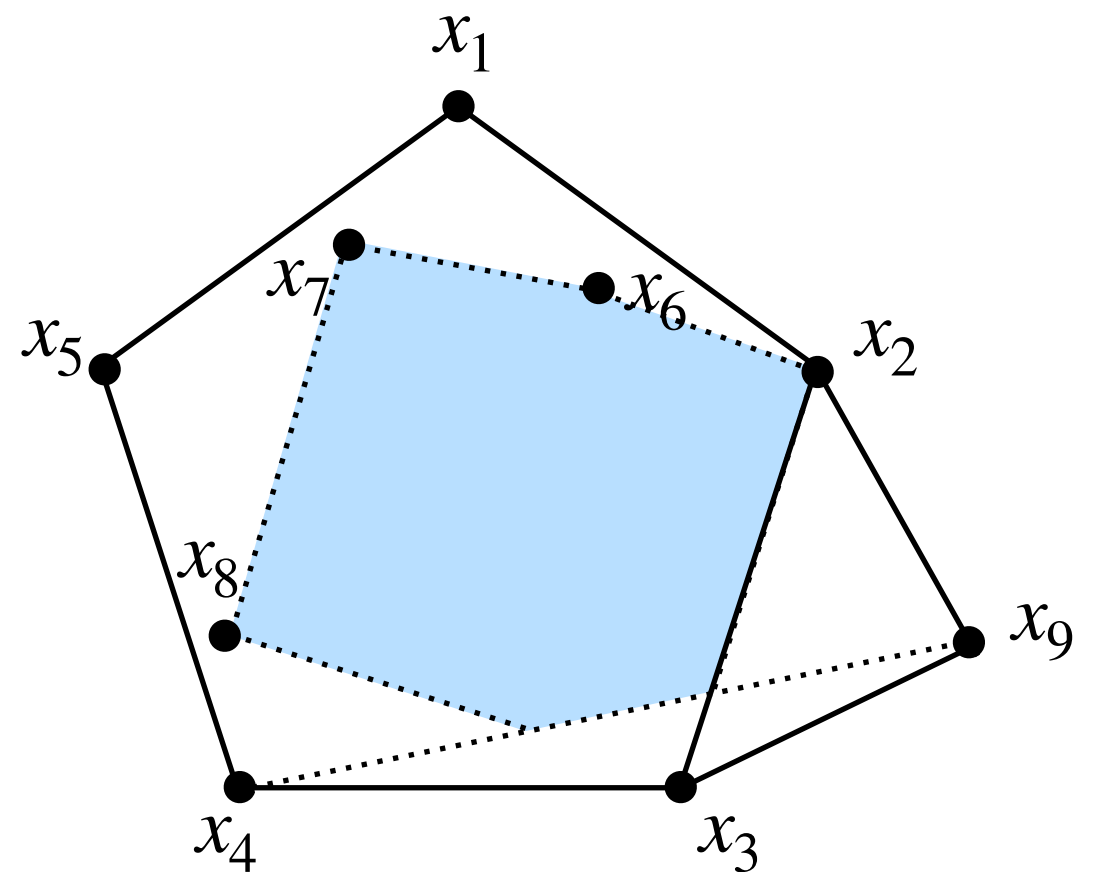
The Creative Process

- Let X_i be an i.i.d. random variable on $\mathbb{R}^d$
- For each $n \in \mathbb{N}$, define the random corpus $C_n = \{X_1, \dots, X_n\}$
- We want to understand $\lim_{n \rightarrow \infty} r_g(C_n)$



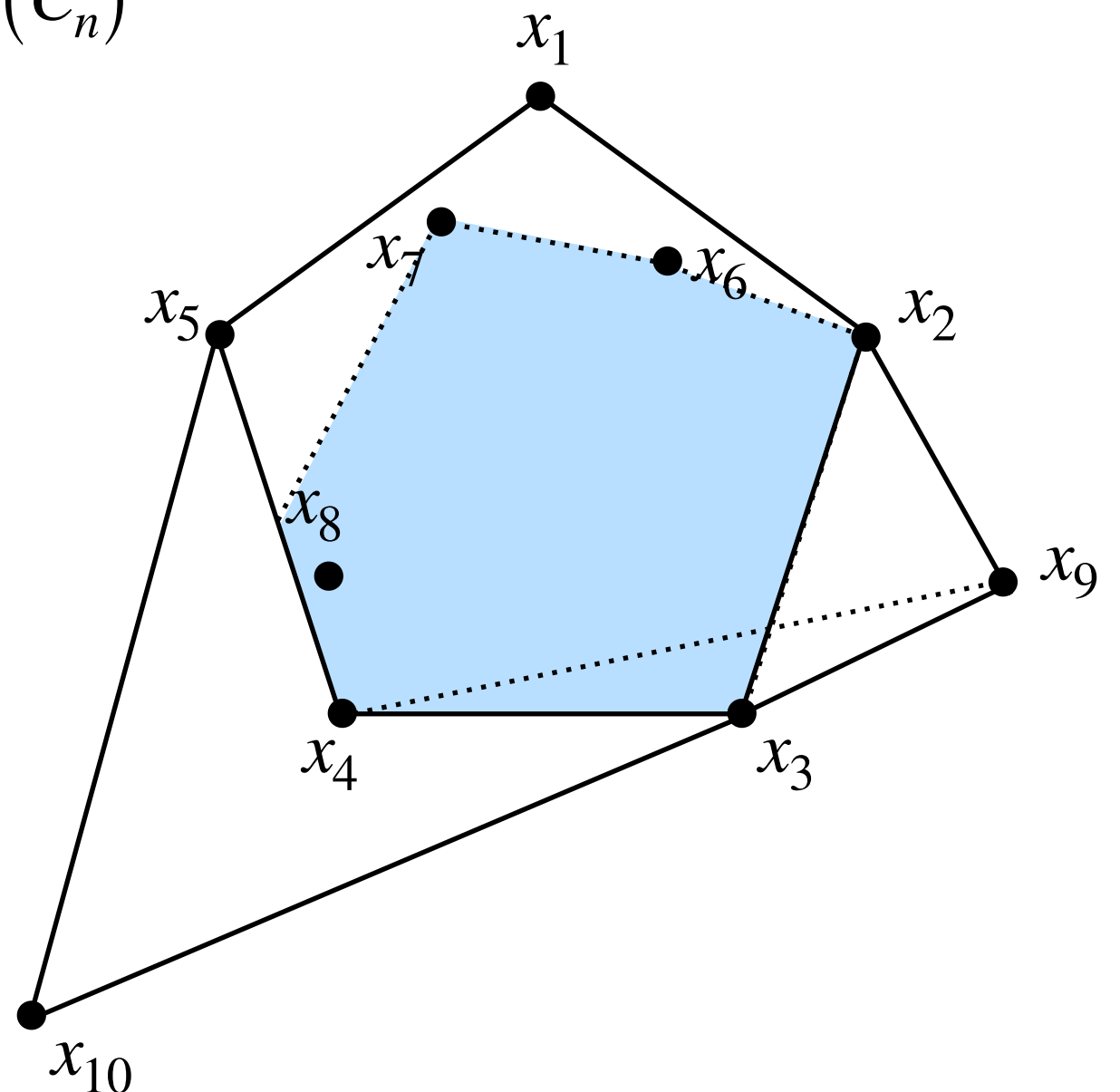
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Light-Tailed Creative Processes

Definition: The random variable $X \in \mathbb{R}$ with CDF F and density f has **light tails** if its hazard rate $h(t) = f(t)/(1 - F(t))$ is nondecreasing.

e.g., Gaussian distribution

Definition (informal): The random vector $X \in \mathbb{R}^d$ has **uniform light tails** if:

1. $\|X\|$ has light tails
2. large draws continue to appear in all directions (uniform lower bound on the conditional density of $X/\|X\|$ given $\|X\| \geq r$, for every $r \geq 0$)

e.g., standard multivariate Gaussian

Light-Tailed Creative Processes

Proposition: Fix a convex-valued generator g . If X_i has uniform light-tails, then

$$\lim_{n \rightarrow \infty} r_g(C_n) = 1 \quad \text{a.s.}$$

- In creative domains with gradual innovation, copyright infringement becomes a vanishing concern as the corpus grows.
- The violation set shrinks to measure zero.

Proof Idea

- Recall that an output is permissible only if it remains generable after removing any one work from the corpus.
- **Lemma:** under uniform light tails, each new draw eventually lies inside a small expansion of the previous convex hull (a.s.).
- Show that deleting **any** existing point shrinks the convex hull only slightly
- The assumption that g is convex-valued allows us to connect this result about the convex hull to a result about the generator
- Once one-point deletion changes $\text{Conv}(C_n)$ only slightly, the same is true for $g(C_n)$

Heavy-Tailed Creative Processes

- Suppose X_i are Pareto distributed on $\mathbb{R}_+$ and let $g = g_{\text{conv}}$ be the convex hull generator.
- **Result:** The limiting permissible ratio $\lim_{n \rightarrow \infty} r_g(C_n)$ is bounded below 1

Takeaways

- In creative fields characterized by sufficiently frequent breakthroughs (such as avant-garde art and literary fiction)
 - protection for individual works may retain its importance even as corpora grow large
 - creators of paradigm-shifting works can continue to have legitimate infringement claims
- By contrast, in creative markets where innovation is largely incremental (possibly including genre fiction and commercial movies),
 - the marginal contribution of any single work diminishes rapidly as the corpus expands
 - generation is eventually unrestricted under our criterion

Generalization to Protected Groups

- In practice, copyright disputes often arise at the level of collections rather than single works



Generalization to Protected Groups

- In practice, copyright disputes often arise at the level of collections rather than single works
- Can generalize our criterion to handle this
- A *collection* of protected works is a family of subsets $\mathcal{A} = \{A_1, \dots, A_n\}$ where each $A_i \subseteq C$ is a **protected group**
 - Our main case: $\mathcal{A}$ includes all singleton sets from the corpus C

Generalization to Protected Groups

Definition:

The **permissible set** is

$$p_g^{\mathcal{A}}(C) := \bigcap_{A \in \mathcal{A}} g(C \setminus A)$$

The **violation set** is

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all works that do not critically
rely on inputs from some $A \in \mathcal{A}$

The **violation set** is

$$v_g^{\mathcal{A}}(C) := g(C) \setminus p_g^{\mathcal{A}}(C).$$

Generalization to Protected Groups

Definition: A collection $\mathcal{A}$ is **stronger** than another collection $\mathcal{B}$ if for every $B \in \mathcal{B}$ there is an $A \in \mathcal{A}$ such that $B \subseteq A$

Result: Fix a generator g and corpus C . If a collection $\mathcal{A}$ is stronger than a collection $\mathcal{B}$ then

$$p_g^{\mathcal{A}}(C) \subseteq p_g^{\mathcal{B}}(C).$$

Superadditivity

Since $\{A \cup B\}$ is stronger than $\{A, B\}$...

Corollary: Fix a generator g and corpus C . For any two protected sets $A, B \in \mathcal{C}$,

$$p_g^{\{A,B\}}(C) \supseteq p_g^{\{A \cup B\}}(C)$$

with strict inclusion possible.

i.e., violations are **superadditive**: the violation of a union can exceed the union of their individual violations

Implication: when multiple creators join a class action, their combined violation set can exceed the sum of their individual claims

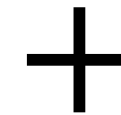
DC Comics



DC Disney



VS.



Disney



LARGER PERMISSIBLE SET

SMALLER PERMISSIBLE SET

Open Questions

- Our paper characterizes the structure of permissible generation but takes as given both the generator and the distribution of creative works.
- A natural extension is to endogenize these objects.
- If creators anticipate how generative systems will use existing works, they may adjust what they produce.

Thank You

Discussion

Restrictiveness:

- Relative to the existing legal criteria, our proposed definition can be either more or less restrictive
- We view our criterion as a complement rather than a substitute for the existing criteria, could be that all should be simultaneously enforced

Implementation:

- Brute-force retraining of an algorithm with the counterfactual training corpus is generally not feasible
- But the emerging literature of machine unlearning develops methods to approximate exactly this (Bourtoule et al., 2019; Xiong et al., 2024)
- Moreover can compare outputs across independently trained algorithms for qualitative evidence

What is the effect of expanding the corpus?

Adding new works to the corpus:

- Expands the set of generable creations (monotonicity of g)
- Introduces new works whose rights must be respected (more constraints)

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Adding new works to the corpus:

- Expands the set of generable creations (monotonicity of g)
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Result: The permissible set is **monotone**:

$$C \subseteq D \implies p_g(C) \subseteq p_g(D)$$

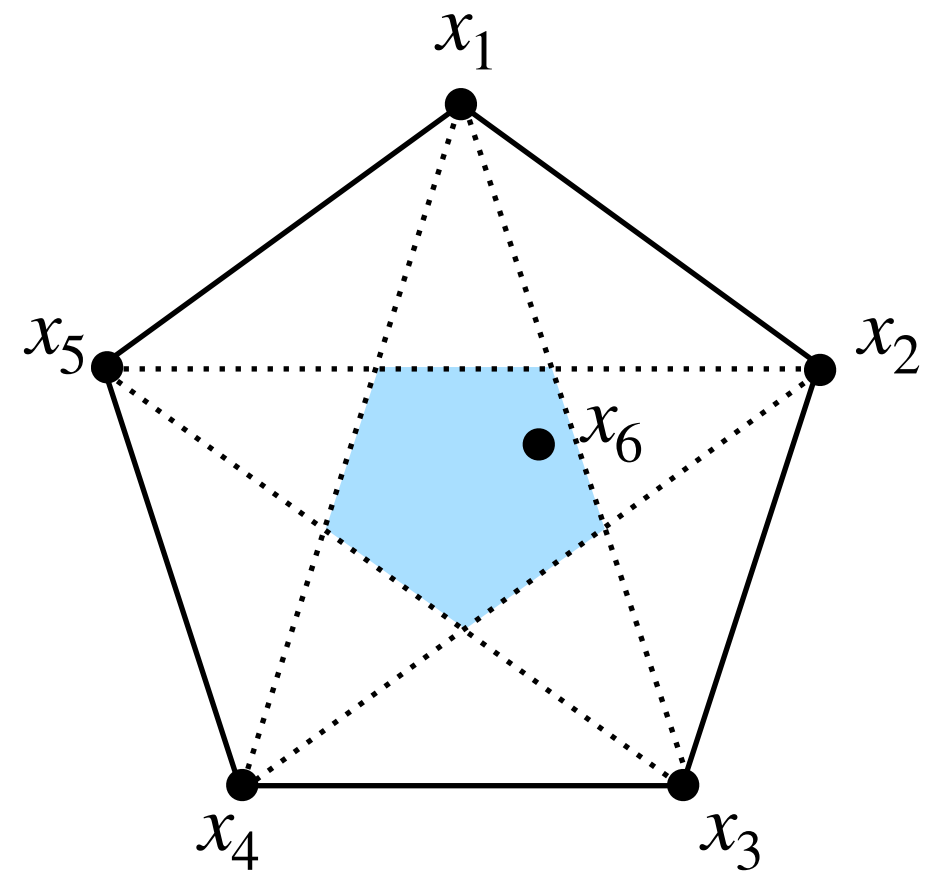
Adding new works to the corpus weakly expands the set of permissible generations.

What is the effect of expanding the corpus?

Result: Moreover, if $D = C \cup \{x\}$ then:

1. If x is permissible, the permissible set is unchanged:

$$x \in p_g(C) \implies p_g(C) = p_g(D)$$



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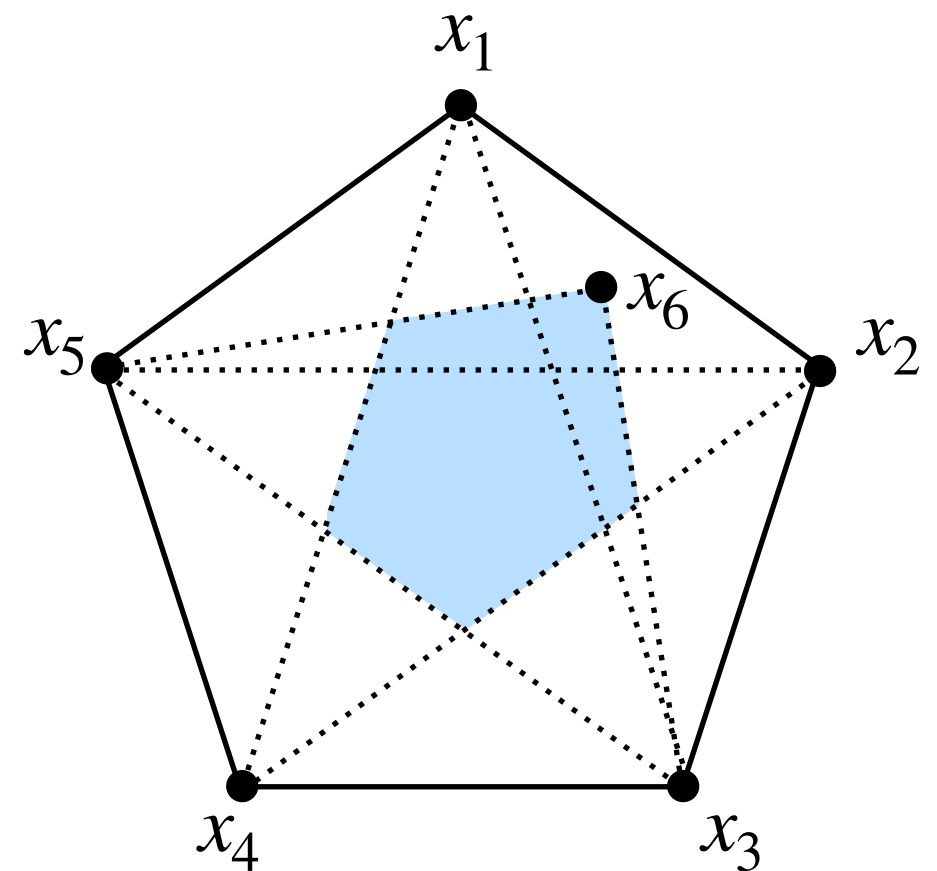
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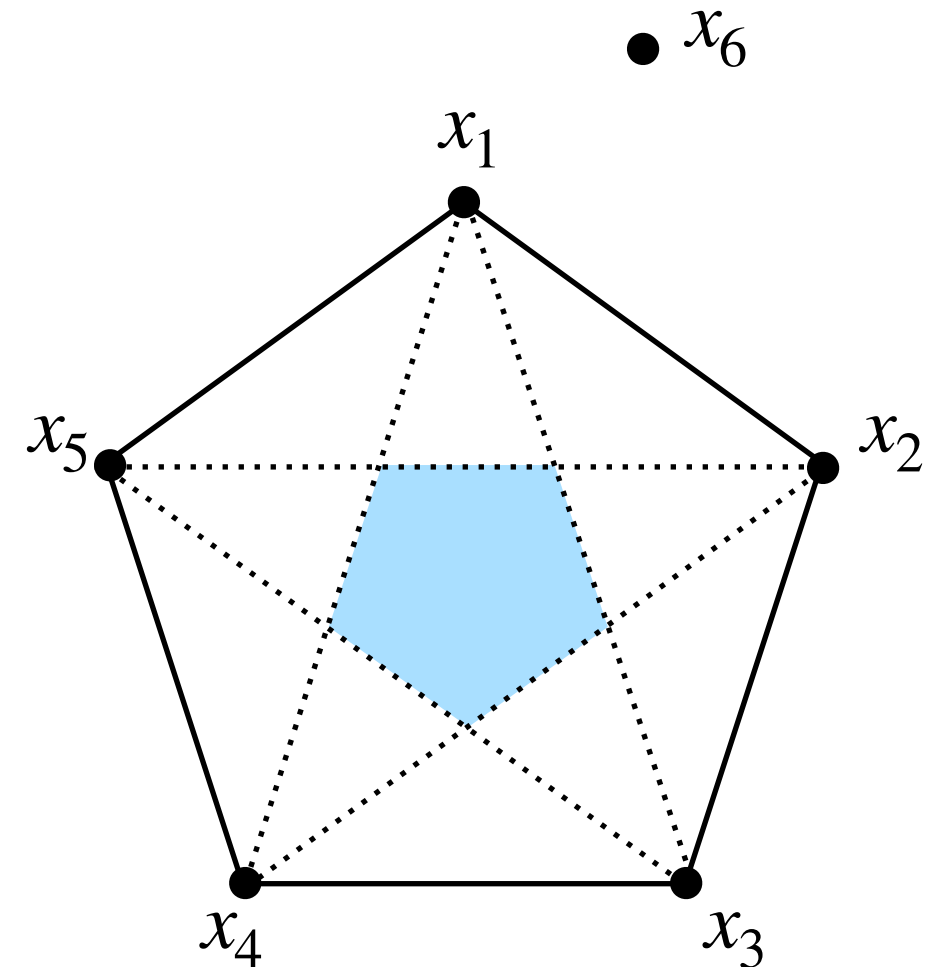
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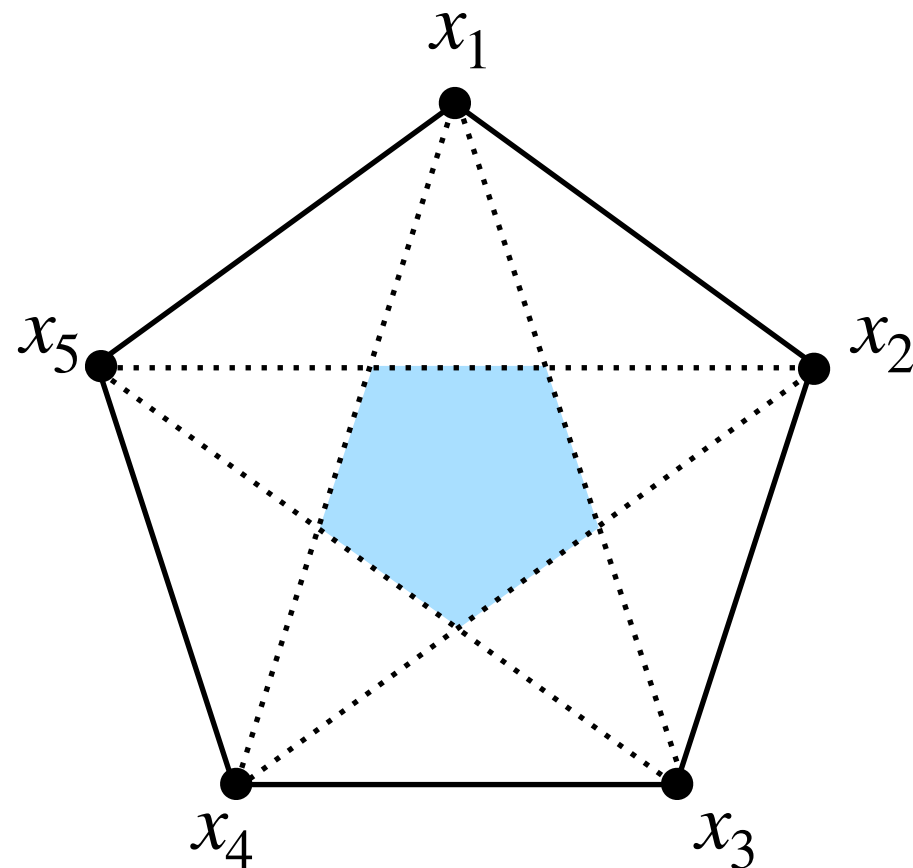
$$x \in v_g(C) \implies p_g(C) \subsetneq p_g(D)$$

3. If x is not generable (i.e., $x \notin g(C)$), then both equality and strict expansion are possible.



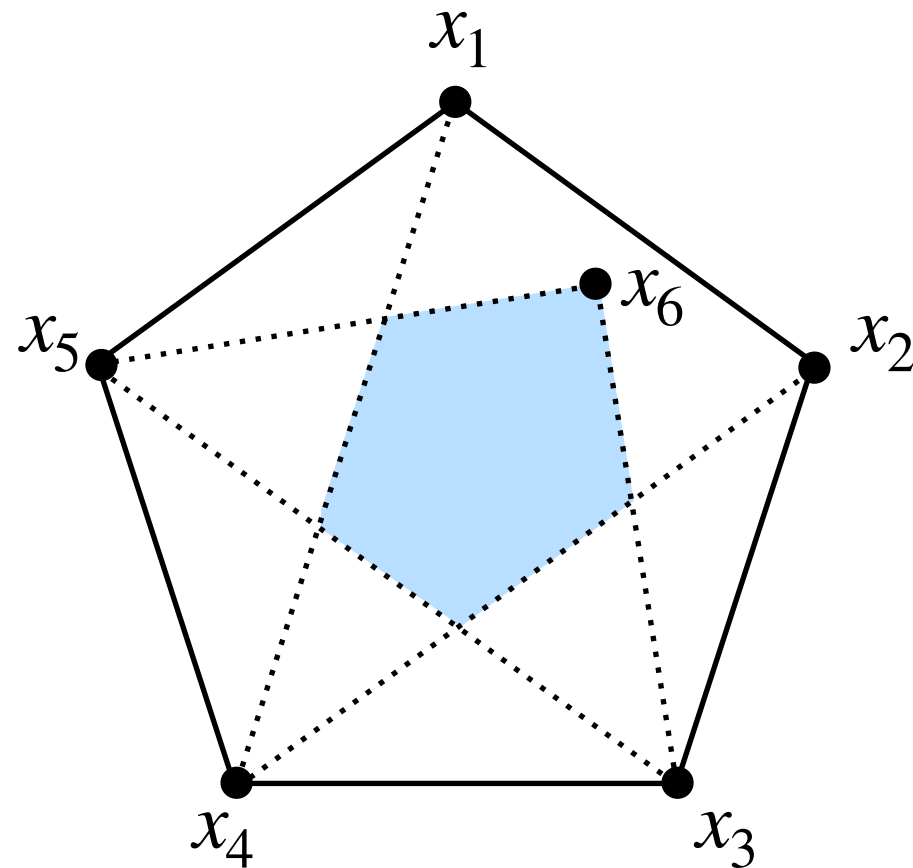
Long-Run Permissibility

Implication: If we continue to add new generations from $g(C)$, eventually everything will be permissible.



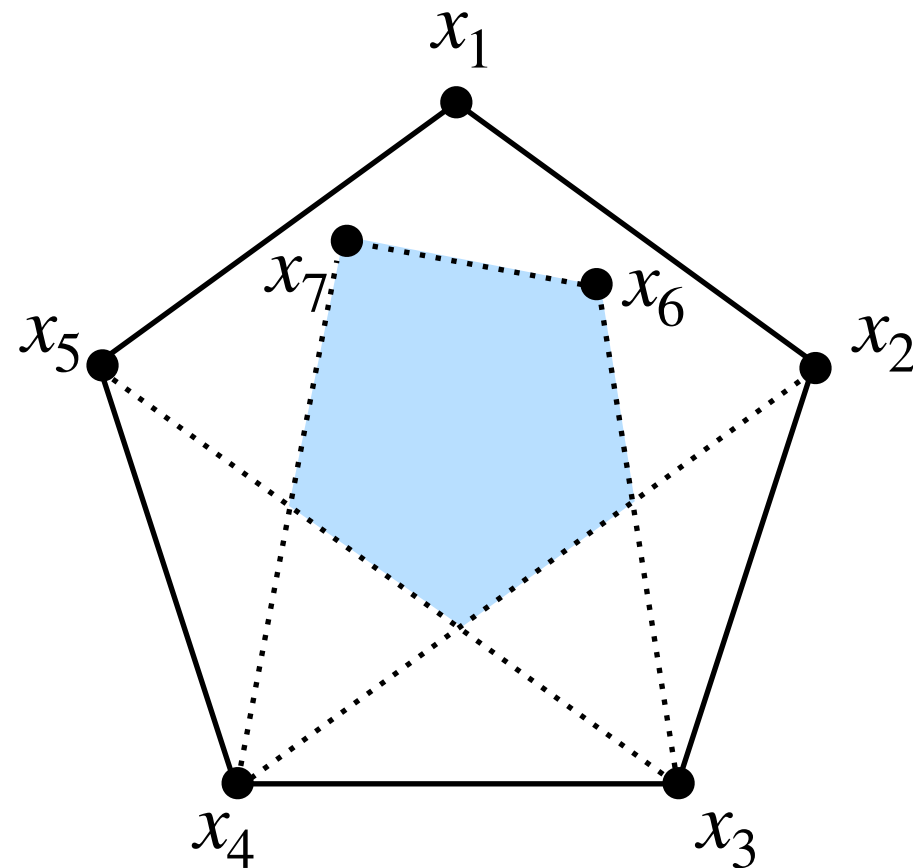
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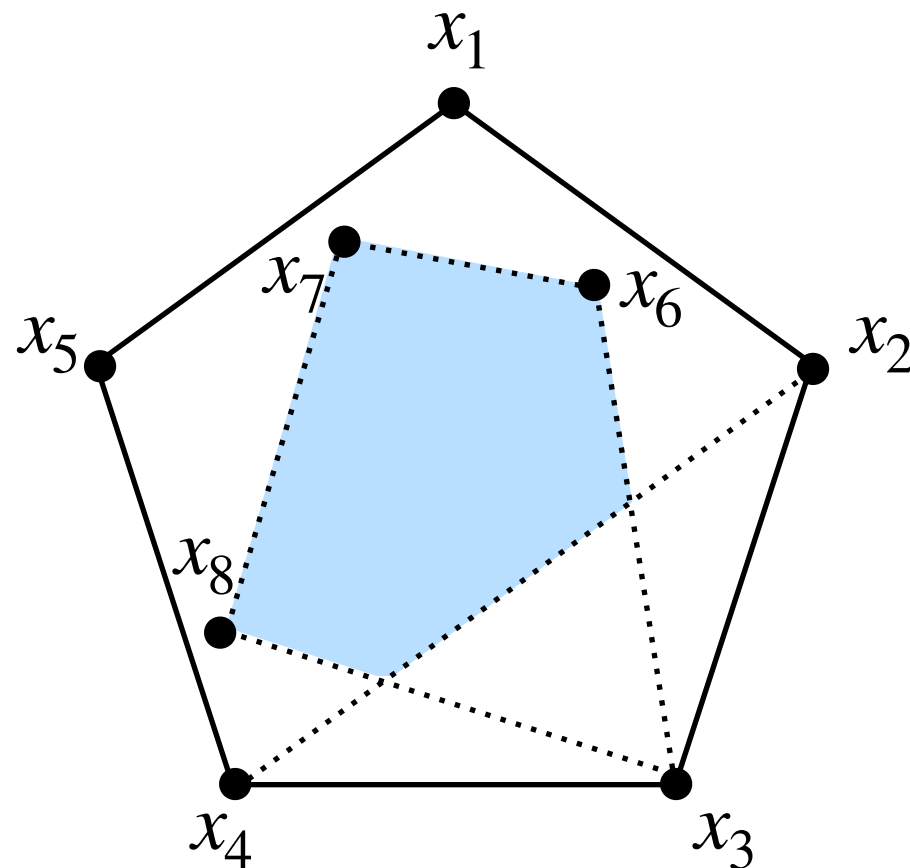
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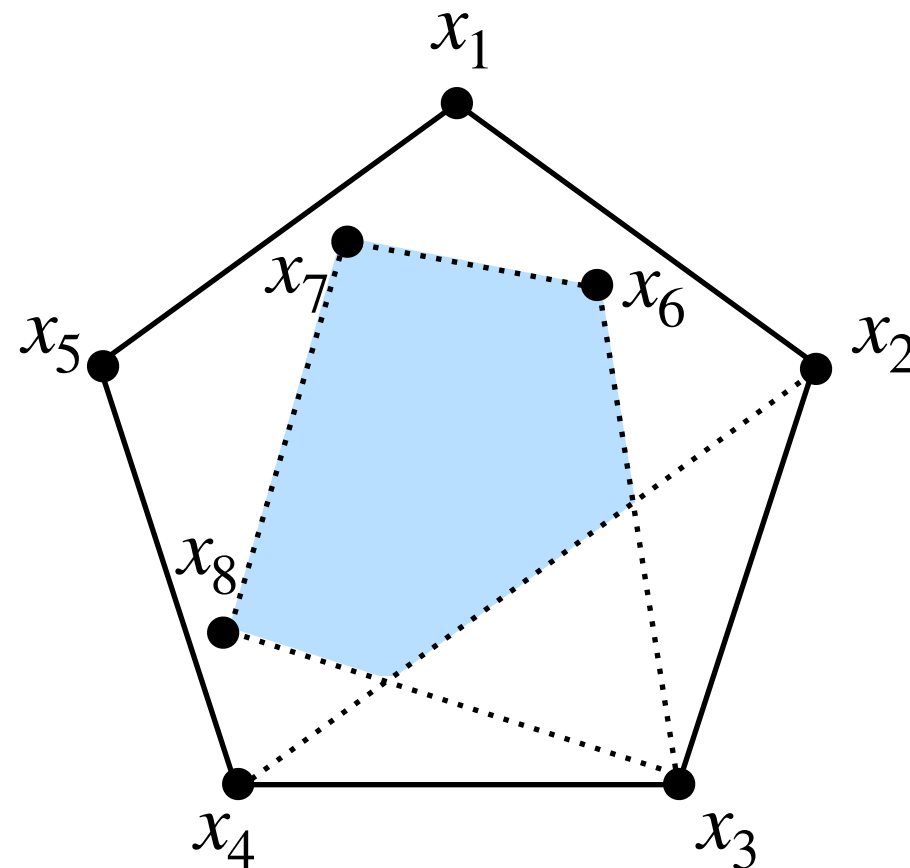
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Long-Run Permissibility

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But in addition, new **human** works are constantly being produced, which may be genuine innovations relative to the original corpus.

role of parametric assumptions

- personalities are uniformly distributed and AI errors follow a multivariate Gaussian distribution
- proof of **theorem 1** ($m_k^* = 2$ for large k) relies heavily on these assumptions, especially the rate of convergence lemmas
- but key statistical phenomenon is more general: in high dimensions,
 - random points isolate from one another
 - the distances between them concentrate around their expectations
- expect generalizations to other distributions (in particular, normal-normal)



two-sided matching

- this paper's model of search is one-sided
- abstracts from the question of whether Subject's best match (once identified) would agree to this pairing
- I do not pursue a complete matching model
- but in many such models, Subject's expected payoff would decrease in the number of candidates considered by Subject's potential partners
- this would further favor in-person search, where each individual searches over fewer candidates



increasingly accurate AI

- main results hold for any quality of AI approximation (i.e., any $\sigma > 0$).
- but would expect that as the AI representations become more accurate, AI-mediated search becomes more (relatively) attractive
- I show that for every fixed number of dimensions k ,

$$d_{k,\sigma}^{AI}(\infty) \rightarrow 0 \text{ as } \sigma \rightarrow 0$$

- i.e., Subject's expected distance to best match on the AI platform converges to zero as the AI's approximation errors vanish
- thus, while the value to increasing the size of the AI platform is capped, the value to increasing the accuracy of the AI clones is not



sequential search

have so far considered a static search problem

consider the following model of sequential search in discrete time

- at $t = 0$, Subject chooses between the in-person regime and AI regime
- in the **in-person regime**, observe X_t in each period $t \geq 1$ and decide whether to stop (and match with best so far) or sample again at cost c_{IP}
- in the **AI regime**,
 - pay an upfront cost κ to participate on the platform
 - in each $t \geq 1$, observe clone distance $\|Y_t - Y_0\|$ and decide whether to stop (and match with closest clone) or sample again at cost c_{AI}

result: Subject prefers the in-person regime when $\kappa > 2c_{IP}$ (i.e., the fixed cost of joining the platform exceeds the cost of meeting two people)



human-AI integration

- one might consider integrating the two approaches as follows:
 - the AI platform simulates interactions between Subject and n individuals using their AI clones
 - recommends to Subject those m candidates whose AI clones are closest to Subject's clone (previously just one)
 - Subject meets these individuals in person and matches to the most compatible among them
- this is **unlikely** to substantially improve Subject's payoffs



human-AI integration

- previously showed that the best match on the AI platform is (for large k) similar to a draw from a uniform distribution over the unit ball
- conjecture that the m best matches on the AI platform are also (for large k) similar to m draws from a uniform distribution on the unit ball
- thus choosing among the AI-sourced candidates is not substantially better than choosing from an equal number of randomly drawn candidates
- suggests that:
 - for any fixed platform entry cost, Subject should optimally choose not to participate on the platform when k is large
 - for large k , Subject should prefer searching over $m + 1$ people in person rather than selecting from m people sourced by the AI platform



intuition

- recall that the selected match i minimizes $\|Y_i^g - Y_0^g\|$
- in a large population, this is an individual for whom that distance is very small
- which group is that extreme draw more likely to come from?
- the smallest distance to any group- g individual, i.e.,

$$M_n^g = \min_{1 \leq i \leq n} \|Y_i^g - Y_0^g\|$$

is asymptotically distributed like an exponential random variable with rate proportional to $f_k^g(0)$ (the density of $Y_i^g - Y_0^g$ at the zero vector)

- so group g 's probability of “winning” is asymptotically determined by the relative size of its exponential rate, and thus by the size of $f_k^g(0)$
- finally, lower noise variance leads to a more sharply peaked density at zero, and hence a higher probability of being chosen

other discussion/extensions (see paper)

- role of parametric assumptions ➡
- two-sided matching ➡
- increasingly accurate AI ➡
- sequential search ➡
- human-AI integrated search ➡

conclusion

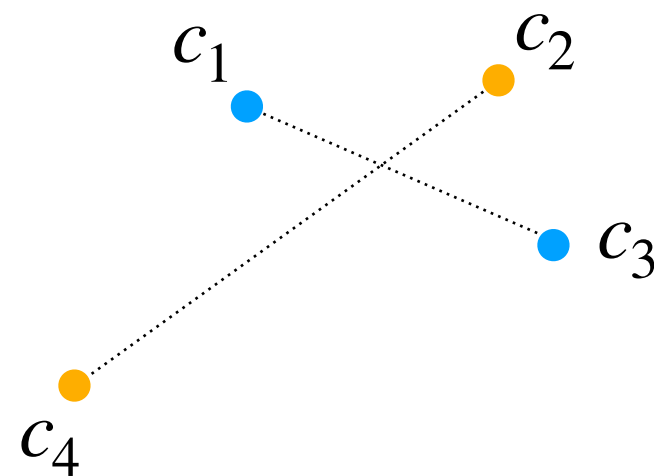
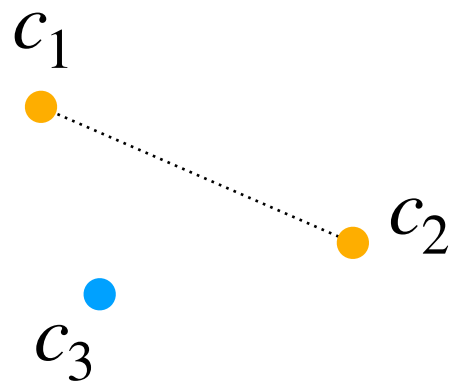
- many papers have compared human and AI evaluation on tasks that are hard for both humans and machines
- but there are important evaluations that are intrinsically easier for people
 - no machine or expert knows better than ourselves whose company we enjoy
- in such settings, human evaluation is necessarily more accurate than the AI evaluation, but also more costly
- can automated search over a sufficiently large number of candidates compensate for the AI's errors during evaluation?
- this paper suggests that the answer depends on the complexity of the target: when sufficiently many attributes are relevant, then the answer is **no**

When is the permissible set non-empty?

Definition: The *Radon number* of a generator g , denoted $R(g)$, is the smallest integer k such that any corpus C of size at least k contains two disjoint and nonempty subsets $A, B \subseteq C$ satisfying:

$$g(A) \cap g(B) \neq \emptyset$$

If no k exists, set $R(g) = \infty$.



Example: for the convex hull generator g_{conv} in $\mathbb{R}^d$, Radon's theorem implies $R(g_{conv}) = d + 2$.

When is the permissible set non-empty?

Result: Fix any generator g . If $|C| \geq R(g)$ then the permissible set is nonempty: $p_g(C) \neq \emptyset$.

Corollary: Fix any convex-valued generator g in $\mathbb{R}^d$. If $|C| \geq d + 2$ then the permissible set is nonempty: $p_g(C) \neq \emptyset$.